

mlx2

Feature & Architecture Review

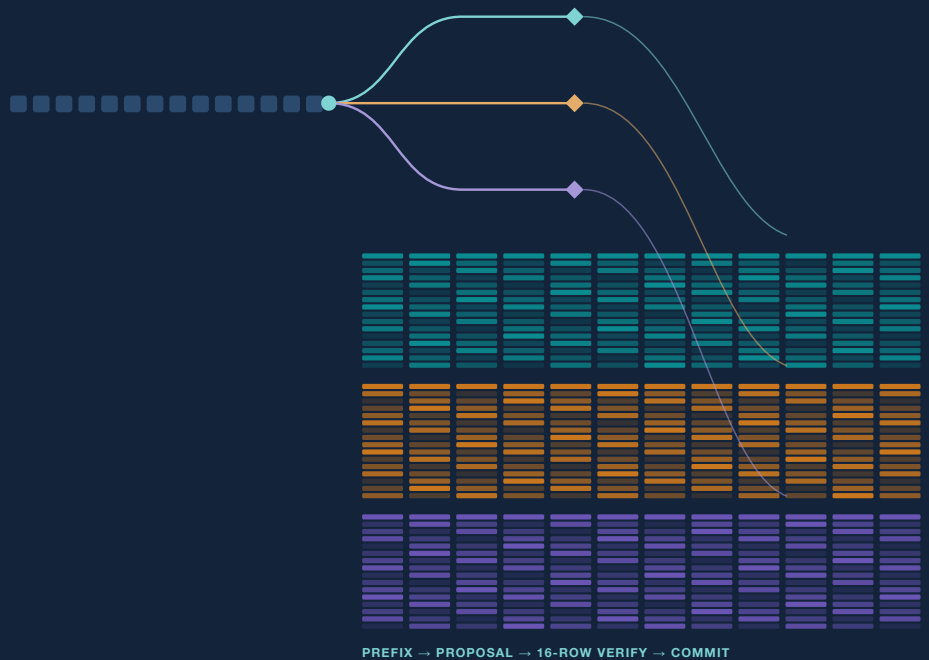
How one request becomes exact, accountable work on unified memory: the adapter boundary, APCv2 for hybrid recurrent models, decode-first scheduling, the speculative family, row geometry, memory, power and thermal control, and what the 6 October sweep changed, measured before and after.

IMPLEMENTED

SELECTED

OBSERVED-USED

QUALIFIED



REVIEW DATE	7 October 2026
SOURCE TREE	main @ 43d561e38 (156 sweep commits on origin/main 187c6aa23)
PUBLISHED	GitHub pierre427/mlx2 8359344 (13 signed topic commits) · private main 43d561e38
GPU EVIDENCE	Before: origin/main 187c6aa23 · Final: main 43d561e38 · M5 Max 128 GB
SUPERSEDES	the 6 October first-round build and the 30 September ReportLab review, same file path

Contents

Mechanism chapters (1–10) are built around diagrams of the real mechanism, with the GPU A/B that tested it alongside; data chapters (11–14) compare the baseline with the final tree, generated from committed evidence.

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Four states, never merged

IMPLEMENTED

Code exists and its tests pass. Says nothing about whether it runs by default.

SELECTED

A flag or an adapter default chose it for a route. A selection can still never execute.

OBSERVED-USED

Run-delta counters and the request receipt show it actually ran.

QUALIFIED

A receipt binds this exact source, native build, artifact and settings, and its gates passed.

Tags used throughout: **DEFAULT** selected by default · **OPT-IN** implemented, off by default · **QUALIFIED** · **APPROXIMATE** changes the output law · **RESEARCH** measured, not selected · **FIXED** changed by the sweep. Every route measured in this document was **unqualified** at its source: any change under `src/` invalidates a receipt, and no receipt exists for the current tree.

The system on one page

From an HTTP request to a terminal receipt, through one model-execution thread, with every refusal and wait explicit.

mlx2 is a single-process inference runtime for Apple silicon. One ServingEngine owns every request from HTTP ingress to its terminal receipt; model adapters own only tensor math, topology and declared defaults. Between the two sit the mechanisms this review explains: an exact prefix and state cache that works for hybrid recurrent models, a decode-first scheduler with a completion-time stall model, a family of exact speculative decoders, row-invariant kernels, and memory, power and qualification accounting that report what actually ran.

104–111 tok/s

Qwen3.8-27B native MTP, B4 aggregate

before 46–51 · now 1.5× ordinary (cohort fix SS-1)

0.44–0.47 s

Flash-Next max decode stall beside a 16K prefill

before 6.9–9.5 s · target 0.5 s

≤ 508 ms

27B neighbour gap p95, every route, beside a 12K prefill

before 548–789 ms · target 500 ms

55–63 tok/s

Prompt lookup B4 aggregate, ordinary 62–63 on the same run

before 25–26 · plain rounds pipelined (82bdcc230, final-pld) · warm 8K TTFT 9.5–9.9 → 1.6 s

Ranges span the repetitions (two, rotated) of each campaign, run through the real HTTP path on an M5 Max. Before: `qualification/runs/sweep-20261006/q38-routes`, `qualification/runs/sweep-20261006/fn` on origin/main 187c6aa23. Final: `qualification/runs/sweep-20261006/final-q38-routes`, `qualification/runs/sweep-20261006/final-fn` on main 43d561e38. Chapter 11 shows every panel; chapter 12 maps each change to its commit.

How to read this review

Chapters 1–10 explain mechanisms. Each is built around a figure that draws the real mechanism with the constants read from source at this tree, and where the sweep measured a mechanism on the GPU, its A/B sits beside the figure. Chapters 11–14 are generated from committed evidence: every number names the campaign directory it came from. Pale bars are **before** (origin/main 187c6aa23); solid bars are **final** (all fixes). Throughout, **implemented**, **selected**, **observed-used** and **qualified** are kept apart (page 2).

What the sweep changed, in one paragraph

The 2026-10-06 sweep ran ten review lanes over origin/main 187c6aa23, measured the baseline on the GPU, fixed what it found, ran a second review-fix round driven by the post-fix GPU reruns, and confirmed the result with final campaigns. It landed 156 commits. The stall bound became a completion-time cost model and now also bounds external-draft and prompt-lookup prefill; co-arriving self-MTP prompts form one cohort; prompt lookup batches on hybrid targets and records recurrent checkpoints; APCv2 charges shared snapshots once without forcing a device sync; NAX serves 6-bit experts; and a power governor now closes the loop on the new telemetry (opt-in). Integration also caught two start-up breaks (an import-order regression and a composition default that refused the DFlash2 tree route), both fixed before the final runs.

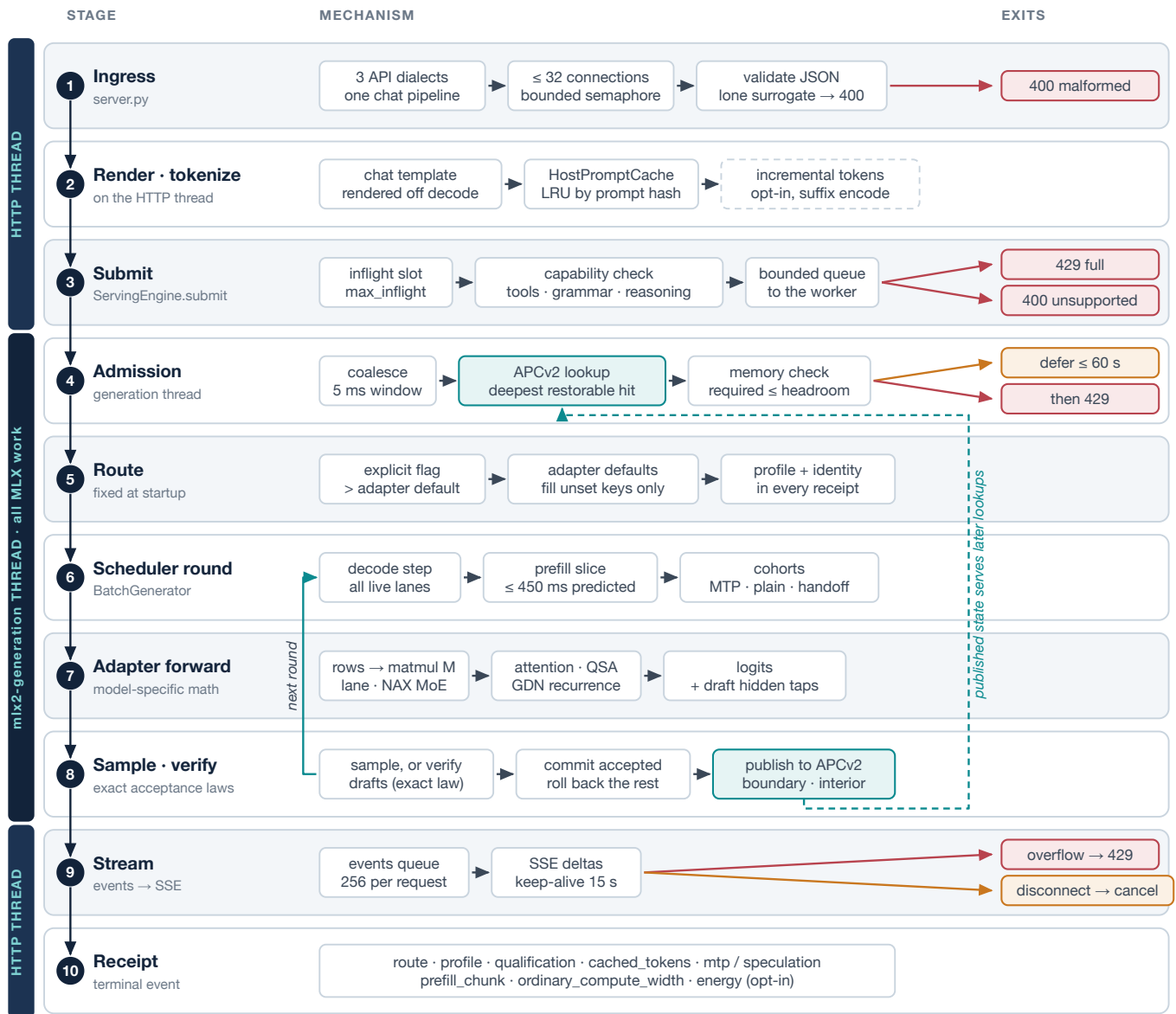


Figure 1. The request lifecycle. Rows are stages in execution order; the left rail shows which thread owns each stage. Rendering and tokenization run on the HTTP thread so a 32K-token template render does not stall decode. Everything that touches MLX runs on one `mlx2-generation` thread: admission, APCv2 lookup and publication, the scheduler round, the adapter forward and verification. Rows 6–8 repeat every round. Failure exits are rose (refusals) or amber (bounded waits). Sources: `server.py` (BoundedHTTPServer, handler_for), `serving.py` (submit, _run, build_batch, receipt assembly), `runtime/generate.py` (BatchGenerator).

Who owns what

Adapters own math and measured preferences; the runtime owns requests, state, scheduling and evidence.

The rule that keeps mlx2 small is that the shared runtime never learns what model it is running. An adapter is the only place a model's tensor math, cache topology and measured preferences live; everything it tells the runtime goes through a declared surface the runtime can check, record and refuse.

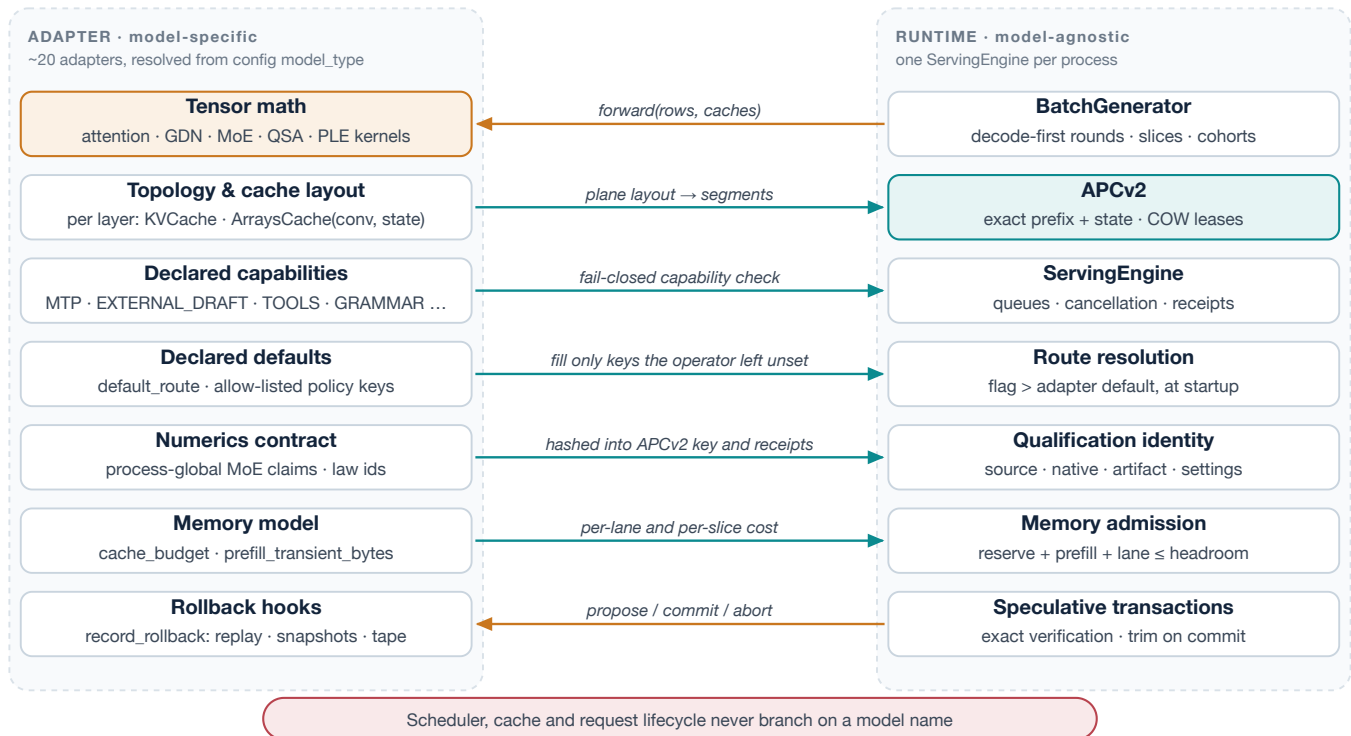


Figure 2. The ownership boundary. Left: what each of the ~20 adapters owns. Right: the model-agnostic runtime. Arrows are the calls and declarations that cross the boundary; teal flows from adapter to runtime, amber from runtime into adapter code.

Sources: `contracts.py` (Capability, StatePlane, Fidelity), `adapters/registry.py` (AdapterResolution, ADAPTER_DEFAULT_POLICY_KEYS), `adapters/process_globals.py`, `serving.py` (apc_semantic_namespace, adapter_prefill_row_tile).

WHAT AN ADAPTER DECLARES

- **Capabilities** from a closed enum; a request needing one the route lacks fails closed.
- **Defaults** read from the class's own namespace, so a subclass never inherits a sibling's measured value (1eb33bb45 fixed exactly that for Nemotron).
- **Numerics contract and process globals**, hashed into the APCv2 key; a conflicting MoE claim refuses the load (e45b862e2).

ROUTE SELECTION HAPPENS ONCE

`resolve_route_selection` picks the route at startup: an explicit flag beats the adapter's `default_route`. Adapter defaults then fill only the policy keys the operator left unset; a default that can never fire at the configured width steps aside and is recorded in `skipped_route_defaults`. An ordinary-decode reference path exists for every family.

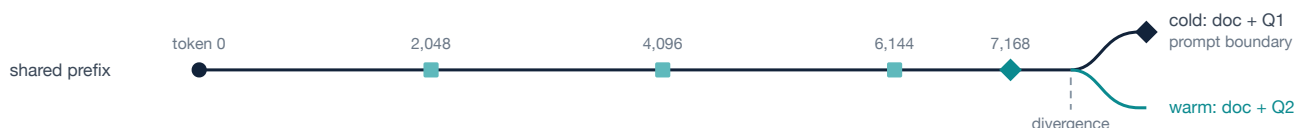
APCv2 for hybrid models

An exact prefix and state cache for models whose recurrent state cannot be trimmed.

For a pure-attention model a prefix cache is easy: KV for every token can be sliced at any position, so any shared prefix is reusable. Qwen3.8, Qwen3.6, Flash-Next and the other hybrids keep a **recurrent** state (Gated DeltaNet) and a short conv state beside their attention layers. That state summarizes every token so far in one tensor; it cannot be cut back to an earlier position. APCv2 is built around that fact.

PREFIX TRIE (one level per token, under one APCKey)

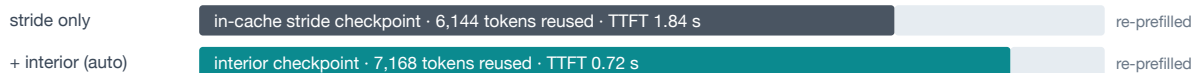
APCKey = model · revision · adapter · tokenizer · cache layout · semantic namespace (every numerical law)



WHAT EACH STATE PLANE CAN RESTORE ALONG THE SHARED PREFIX



WARM HIT = DEEPEST POSITION WHERE EVERY PLANE IS RESTORABLE (measured)



■ recurrent checkpoint kept inside the lane's cache (stride 2048, at most 4)

◆ interior checkpoint published to APCv2

ONE APCv2 ENTRY

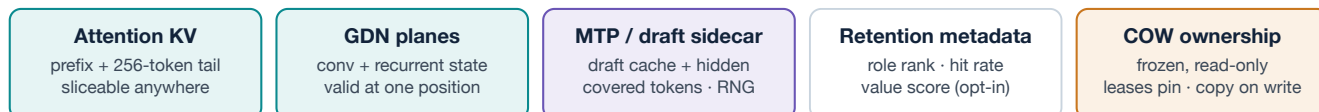


Figure 3.1. Why interior checkpoints matter on a hybrid. Two requests share an ~8K-token document and differ in the question. KV could serve the whole shared prefix, but the recurrent planes exist only where a state was captured: the in-cache stride checkpoints (MLX_LM_STATE_CHECKPOINT_STRIDE=2048, at most 4 kept) and, when enabled, an interior checkpoint published to APCv2. Measured on Qwen3.8-27B ordinary (qualification/runs/sweep-20261006/q38-features, warm request of each pair, two reps): the hit lands at the deepest position where every plane is restorable. Positions 2,048/4,096/6,144/7,168 are the observed cached-token counts and stride; the divergence point is schematic.

HOW A LOOKUP DECIDES

The index is a per-token trie under an `APCKey` (model, revision, adapter, tokenizer, cache layout and a semantic namespace that nests every numerical law: adapter numerics, prefill execution, lane matmul, int8 prefill, weight streaming, and the recurrent-state codec outermost). A lookup returns the exact entry, the deepest stored prefix and the shortest extension. On a hybrid, a longer entry cannot be branched at the divergence (`untrimmable_branch`), so the usable hit is the deepest entry whose planes all sit at or before the shared prefix. Draft routes additionally require a sidecar; since f6a160933 the deepest sidecar-carrying ancestor is served instead of a full miss (APC-1).

WHAT GETS PUBLISHED

Every request publishes its prompt boundary (P-1) and its finished lane. With `apc_interior_checkpoints: auto` (an adapter default on the Flash-Next, Qwen3.8-27B and Qwen3.6-35B native-MTP routes) up to 4 interior positions are planned (turn markers, then a tail lattice with stride ≥ 256), budgeted against half the execution headroom, captured during prefill (slices end exactly on them) and published when the prompt finishes. Approximate state never enters: the only publish path skips approximate lanes and quantized planes.

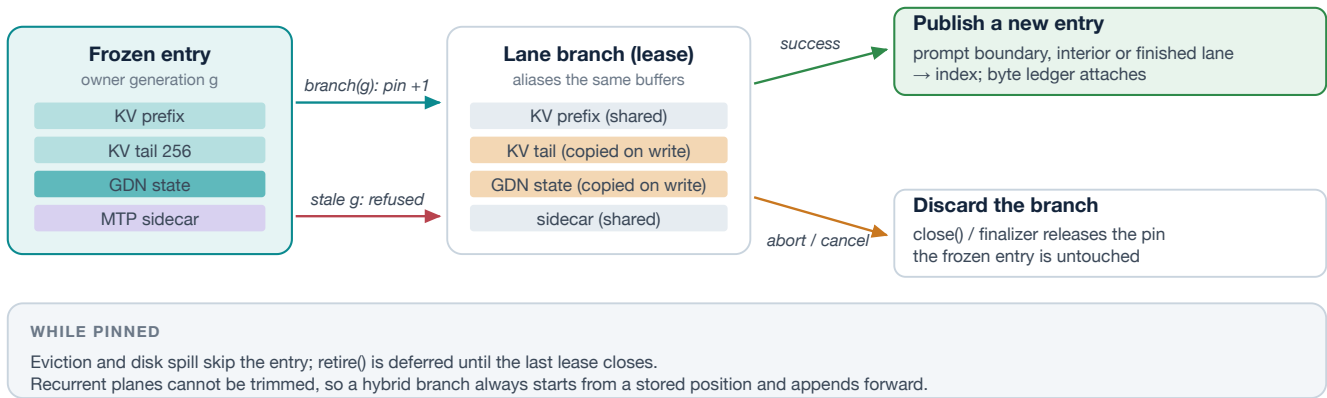


Figure 3.2. Copy-on-write branches and leases. Freezing aliases MLX buffers without copying; a branch checks the owner generation, pins the entry and copies a segment only on its first write (KV is split into a prefix and a 256-token tail so appends copy little). Source: `runtime/cow_cache.py` (`freeze_prompt_cache`, `COWCacheOwner.branch`), `runtime/apc_v2.py` (`_store_locked`, `retire`).

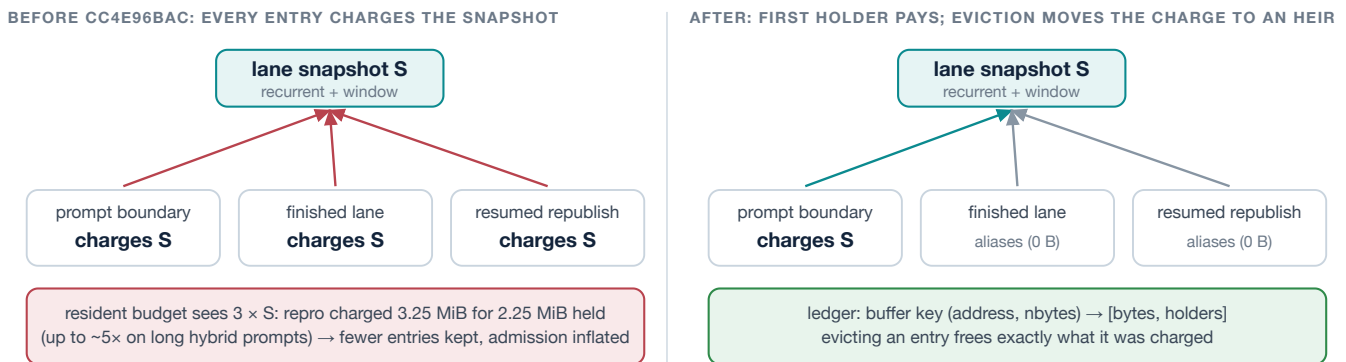


Figure 3.3. The shared-snapshot byte ledger (COW-1, fixed in cc4e96bac). A lane keeps its restore snapshots across every entry it publishes, so several entries alias the same buffers. Each entry used to charge them again; the ledger now keys buffers by (address, nbytes), charges the first holder and moves the charge to an heir on eviction. Repro numbers from the commit message. The first ledger keyed buffers by data address, which evaluated the pending snapshots and forced a device sync on every APCv2 store during prefill; b02360126 keys by an ownership token carried by every alias instead.

OPT-IN: VALUE-AWARE RETENTION (F44F355B5)

The default eviction order is categorical rank then recency: rolling < interior < default/junction < prompt boundary, with an interior checkpoint that has been hit promoted one rank. That keeps one-off prompt boundaries alive while shared tool-schema and document checkpoints are evicted first. `apc_retention_policy: "value"` instead evicts the entry with the least saved prefill per freed byte: $\text{value} = \text{rate} \times (C(d) - C(a)) / \text{unique_bytes}$, with $C(d) = d + d^2 / (2 \cdot 8192)$, `rate` a hit rate decaying with a 600 s half-life (role priors before the first hit), `d` the depth it serves and `a` the deepest other entry on its path. Unique bytes come from the ledger above. APC-2 (cd9fb13b3) makes the hit counts it relies on per request, and withdraws hits the route then refused.

Scheduling

Decode first, then the largest prefill slice the completion-time model predicts will fit the stall target.

The scheduler (`BatchGenerator` in `runtime/generate.py`) runs rounds. Each round steps every live decode lane once, then may run one prefill slice for waiting prompts. The design question is how big that slice may be: too small and long prompts take forever (and small-M forwards cost 2–3× more per row); too big and every decoding user sees a long pause.



Figure 4.1. A decode-first round. Lanes A and B decode together; prompt C's 12K tokens are sliced so that A's token gap stays within the stall target; timing is schematic. Defaults: target 500 ms held 10% back (450 ms); slices round down to a 64-row grid (an adapter may declare its own tile, 48289d962; none does yet); contended prefill time × fair share 0.5 becomes debt, and a round with debt skips prefill so decode repays it; under SRPT (native MTP, ≥ 4 lanes) the shortest remaining prompt goes first, bypassing at most 3 times. External-draft and prompt-lookup routes apply the same bound with fair share 0, because they already interleave one slice per decode poll (673aa9fe0). Source: `runtime/adaptive_policy.py` (DecodeTimeFairness, PrefillOrder).

From a running maximum to a completion-time model

Before the sweep, the stall bound used the **fastest** prefill rate ever observed, per row. One fast sample pinned it after a slowdown, a B-row forward cost about B times the target, and ordinary, mixed and self-MTP samples were pooled (SS-2). Commit d28a774aa replaced it with `PrefillCostModel`: $t = a + b \cdot R$, where R is the forward's actual rows (prompt rows × padded width, plus decode rows in a mixed forward), a is the median recent decode step (clamped to half the cheapest prefill sample) and b is the **90th percentile** per-row cost over the last 16 samples, kept separately per kind (ordinary, mixed, MTP) and KV-depth bucket (4K / 16K / 64K). A slice 3× slower than predicted (or 1.5× twice) drops all but the latest sample at once; a faster regime needs four consecutive fast slices. Bounds never grow past 2× the largest forward seen. The review-fix round added three refinements: with no sample at all the model probes one 64-row tile instead of a 512-row guess (about 650 ms on the 27B; 7502a23ca), stale samples expire when a bound is asked for (8c7e1fe78), and the estimator is part of the route identity (acadd8b76).

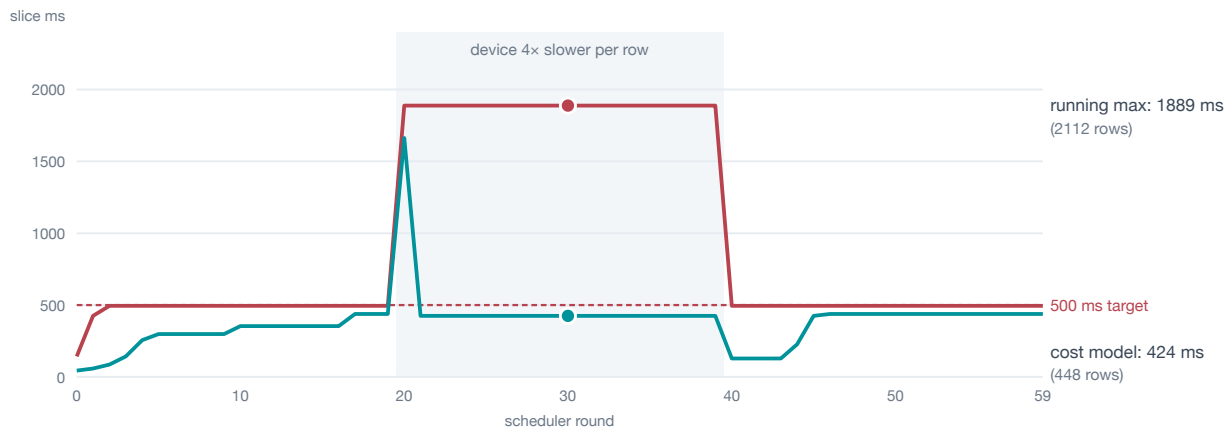


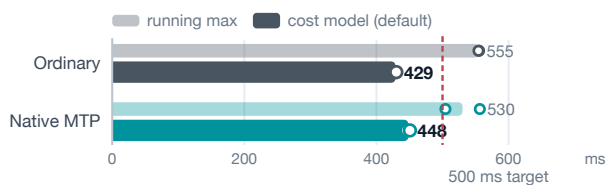
Figure 4.2. The two estimators under the same slowdown, run live. At build time this document loads the real `DecodeTimeFairness` class from `src/mlx2/runtime/adaptive_policy.py` twice (`estimator="running_max"` and the default `"cost_model"`) and drives each in closed loop against a synthetic device ($t = 30 \text{ ms} + 0.22 \text{ ms/row}$, 4× slower per row in rounds 20–39). The running maximum keeps its fast-regime slice and overshoots to 1889 ms; the cost model pays one slow slice (1663 ms in round 20), then re-sizes to 448 rows (424 ms). When the device speeds up again the model needs four consecutive fast slices before it grows the slice (rounds 40–44). The device is synthetic; the estimator code is not.

The same A/B on the GPU

The estimator is a policy key, so the two laws can run on identical code. On Qwen3.8-27B with a 12K prompt arriving beside a decoding lane, the cost model moved the neighbour's p95 gap –23% (ordinary) and –16% (native MTP) and its maximum –18% and –13%, bringing every p95 under the 500 ms target; the price is the long prompt's own TTFT (+1% and +6%), because its slices are smaller. Without contention (B4) the two are equivalent. Medians of two reps; changes are median to median.

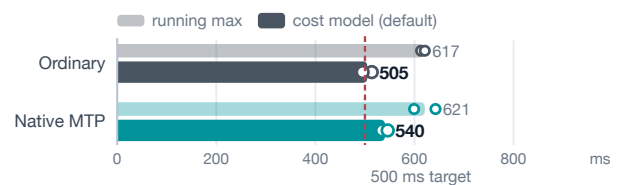
Neighbour gap p95 beside a 12K prefill

Qwen3.8-27B, ms · lower is better



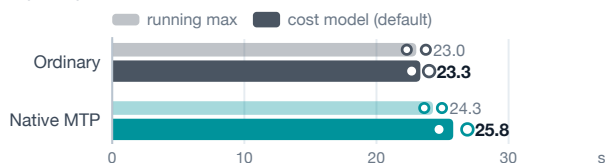
Neighbour gap max

same window, ms · lower is better



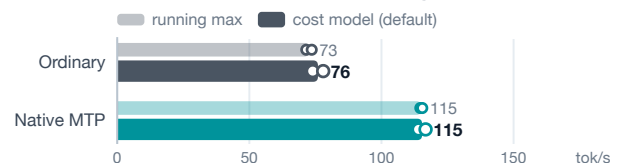
Long-prompt TTFT under contention

the 12K prompt's first token, s · lower is better



B4 aggregate decode

no contention: the estimator should not matter · higher is better



Evidence: `qualification/runs/sweep-20261006/estimator` · intermediate integration tree 816aef02c; both arms ran the same code, differing only in `decode_fairness.estimator`.

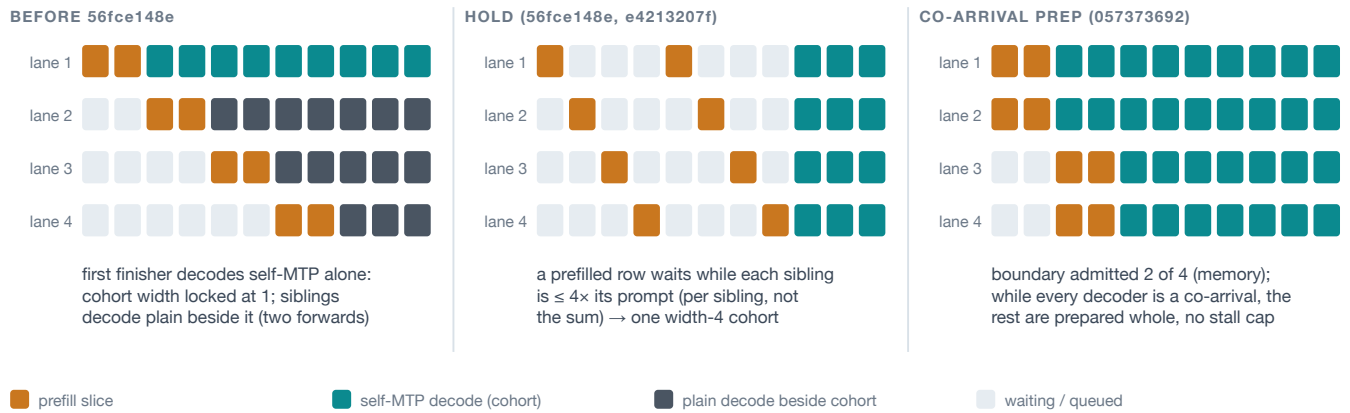


Figure 4.3. Cohort formation for self-MTP. Left: the baseline lock (SS-1). Middle: the co-arrival hold (56fce148e), bounded per sibling since e4213207f so an eight-prompt burst is held like a four-prompt one. Right: when a first idle boundary admits only part of a burst (memory admission), the rest are prepared whole while every decoding lane is one of their co-arrivals (057373692); there is no earlier neighbour to protect. With 27B prompts autoscaled to 512-row steps, any prompt over 513 tokens is multi-chunk, so concurrent chats hit this. Native MTP B4 went from below ordinary to 1.5 $\times$ ordinary (chapter 11).

ALSO IN THE SCHEDULER THIS SWEEP

- **Self-MTP preparation** is timed as `kind=mtp`, owes debt and is cut to the stall budget, oldest row kept (SS-3); each one-call preparation is charged its own fixed cost (470e7d3af).
- **Autoscale** sizes warm requests from history + new tokens everywhere (2d4a9a676, SS-4); **batch geometry** beside SRPT now fails closed (4a0c00bae).
- **Gap attribution**: every contended decode interval is split into forward time and other time, so a max-gap outlier can be traced (d0ac480ae).
- **GPU keep-warm** (opt-in, 726837c12) ticks a one-element add every 0.5 s after work. Measured wake penalty on the M5 Max is about 0.3 s after 20 s idle, not the ~1.7 s claimed elsewhere, so it stays off.

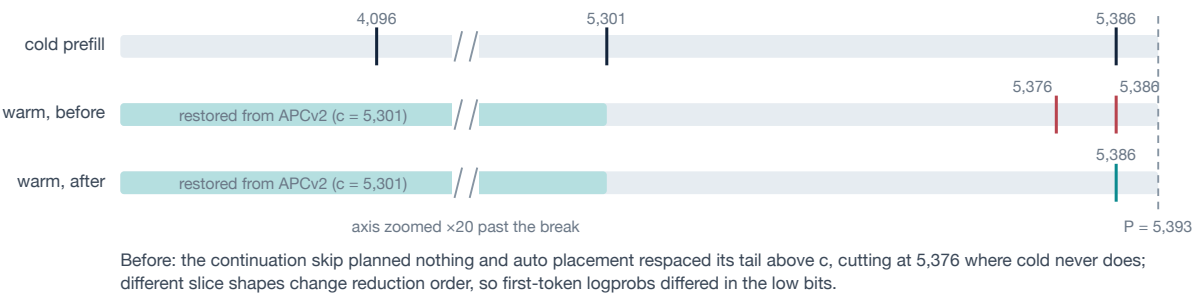
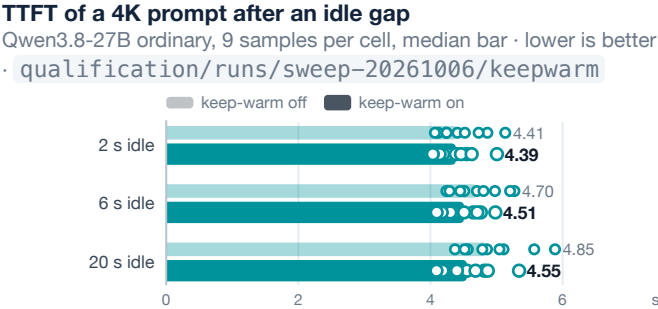


Figure 4.4. Warm slices aligned to cold cuts (APC-3, d23b16895). Warm capture positions are restricted to the cold plan's cuts; receipts carry `slice_aligned`. Positions from the lane repro.

The speculative family

Four ways to propose, one way to verify: exact laws, packed rows, per-plane rollback.

Every speculative route in mlx2 has the same shape: something cheap proposes tokens, one target forward scores the anchor plus all proposals, an exact acceptance law decides how many survive, and the state of every plane is committed for the accepted prefix and rolled back for the rest. The routes differ only in where proposals come from and how rows are packed.

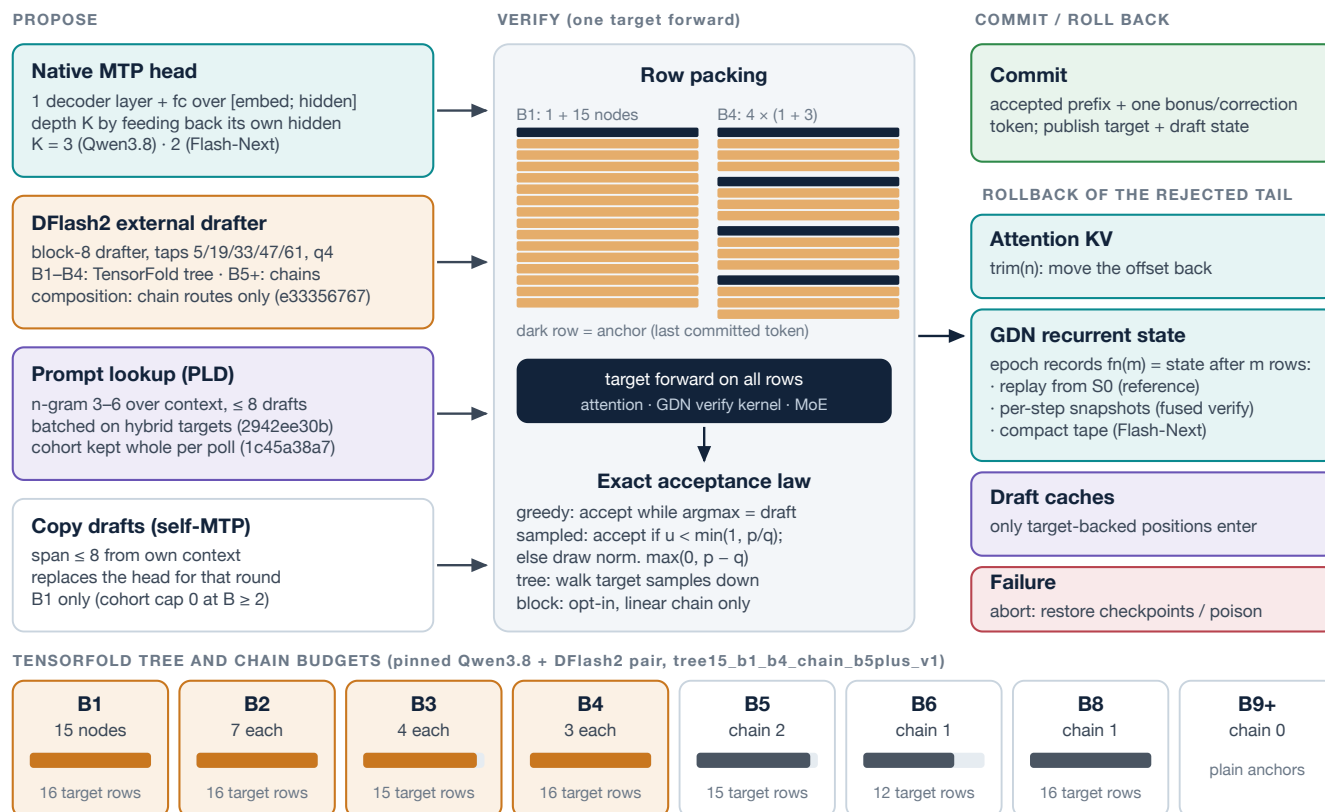


Figure 5. The speculative family on one page. Left: the four proposal sources. Middle: packing into target rows and the acceptance laws. Right: what commit and rollback do per state plane. Bottom: the default

`tree15_b1_b4_chain_b5plus_v1` budgets on the pinned Qwen3.8 + DFlash2 pair, which hold about 16 target rows per round at every width (1f9c22d1d moved B5+ chains from 8 rows per lane to `rows // lanes - 1` drafts). Sources: `hybrid_speculative.py`, `external_speculative.py`, `speculative_sampling.py`, `pld.py`, `copy_draft.py`, `adapters/qwen38_27b.py`. KV rollback is an offset move; a GDN layer registers per verify forward a function that returns the exact state after m rows (replay, per-step snapshots or Flash-Next's compact tape), and a rollback beyond the recorded window (64) fails loudly rather than guessing.

Exactness is proven, not assumed

The spec lane built an oracle that enumerates every RNG branch of the real verifier code symbolically and compares the completed output law with the target joint law: every production verifier matched to about $1e-12$ (float64) or $2e-5$ (float32 MLX laws), and the oracle catches two classic wrong verifiers. It now lives in `tests/` (34216accb), which also

fixed the one measure-zero hole it found: a rejection that leaves no residual mass now draws from the target law instead of raising (SPEC-07). The review round tightened it further: the symbolic uniform now refuses non-strict comparisons, which found that block and self-MTP verification accepted on $u = h$ (a forbidden draft on an exact-zero draw; now $u < h$, 8ca62ebf5), and a hybrid abort now rewinds recurrent rows as well as KV, with failed prompt-lookup rounds rebuilt from their checkpoints (1d25dd7d1). In the final B1 short-prompt runs the DFlash2 tree accepted 324 of 918 proposed nodes (35%; a tree proposes 15 nodes per round but at most one path can be accepted); on the copy workload 890 of 1,140 (qualification/runs/sweep-20261006/final-q38-routes).

Block verification (Sun et al. 2024, arXiv:2403.10444) is implemented for the linear external chain (ccbf2a99c): it draws all K uniforms up front and keeps the last position whose test passes, which can accept more than token-wise verification under the same output law. The gain exists only for stochastic proposal laws. It is **OPT-IN**, **UNQUALIFIED**, and no GPU A/B has run (provenance/block-verification.json).

The four sources compared

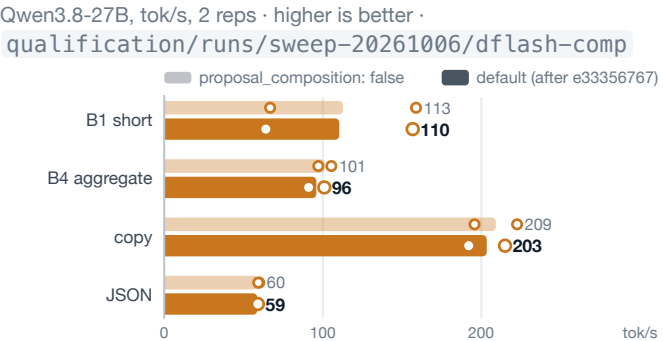
SOURCE	SELECTED WHERE	DRAFTS PER ROUND	AT B > 1	WARM APCV2 REUSE
Native MTP	default route with a head: Qwen3.8-27B, Flash-Next, Xing, Qwen3.6-35B	K = 3 / 2 / 1 (cap 3)	one true-batched cohort; ordinary handoff above width	yes, with the draft sidecar; hit depth unstable (ch. 14)
DFlash2 tree	explicit external route, pinned Qwen3.8 pair	15 nodes at B1 (7, 4, 3 at B2-B4)	packed TensorFold B2-B4; chains B5+	no: no draft state below the prompt boundary (by design)
Prompt lookup	opt-in route	up to 8, adaptive latch	batched since 2942ee30b, cohort whole since 1c45a38a7; plain rounds pipelined since 82bdcc230 (B4 at parity)	yes since cb26c206b (checkpoints recorded)
Copy drafts	inside self-MTP, Qwen3.8 and Flash-Next defaults	span ≤ 8, point-mass q	refused (cohort cap 0), reported as such	rides the MTP route

DEFAULTS, STATED PRECISELY

Native MTP is the default route where the artifact has a head. The DFlash2 tree is the default of the explicit external route for one pinned revision pair, not the model's default. Prompt lookup is opt-in; copy drafts run at B1 and are refused in cohorts.

Proposal composition became a default on main (66b2f9d1e) for every external drafter with draft distributions, including the tree pair, which refuses it: the default DFlash2 route failed at start-up. e33356767 limits the default to **chain routes** (Muse DFlash2, North EAGLE: unmeasured, unqualified) and lets the tree route opt out explicitly. The rerun at right shows the two equivalent on the tree route.

DFlash2 tree route: default vs composition off



Row geometry and kernels

How decode, verify and prefill rows become tiles, and why 15, 16 and 17 rows are different machines.

A transformer forward is mostly matrix multiplies whose M dimension is the number of rows in flight. Decode contributes one row per lane, verification K+1 per lane or a 16-row tree, prefill hundreds to thousands. Which kernel runs, and therefore which bits come out, depends on M. mlx2 makes that dependence explicit instead of accidental.

WHERE ROWS COME FROM

ordinary decode

B lanes $\rightarrow$ B rows

4 rows

self-MTP verify

B $\times$ (K + 1) rows

8 rows

tree verify (B1)

1 anchor + 15 nodes

16 rows

Flash-Next verify

proven bit-exact ≤ 17

17 rows

prefill slice

64 ... 8192 rows per forward

prompt rows $\times$ padded width

16-ROW FRAGMENTS ON THE M5 TENSOR UNIT

M = 15

1 fragment, 1 idle row

M = 16

1 full fragment

M = 17

2 fragments, 15 idle rows

Tree GDN state uses a 16-slot variant up to 16 rows, then 32 slots.

WHY THE KERNEL CHOICE CHANGES BITS

Stock MLX quantized matmul

kernel and summation order chosen by M
 $\rightarrow$ the same row can round differently at B1 vs B4
 per-row cost climbs from 4 rows on M5

Lane matmul (row-invariant small-M, M5)

split-K fixed by weight shape and format only
 $\rightarrow$ every row gets the same bits at any M
 grouped q-k-v / gate-up share one stacked launch

DEFAULT LAW lane-matmul-v1+stock-below[bf16:16, fp16:16, q2...q8:8]+grouped $\cdot$ stock above 32 rows $\cdot$ MoE: off

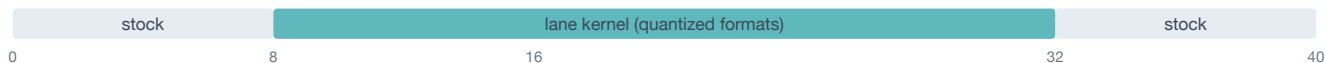


Figure 6. Rows to tiles. Top left: where rows come from. Top right: the M5 tensor-unit fragment is 16 rows (lane matmul `ROW_BLOCK = 16`; TensorFold pads to `16 * [M/16]`), so the default tree's 16 rows fill one fragment exactly while 17 rows need two. The tree GDN kernel likewise has a 16-slot variant and falls to 32 slots above 16. Bottom: why lane matmul exists and its default law. Sources: `runtime/lane/matmul.py`, `lane/policy.py`, `lane/installer.py`, `external_speculative.py` (`_chain_draft_cap`), `qwen38_tree_gdn.py`, `qwen4_fused_gdn_verify.py`.

Under the default `crossover` mode, rows below the per-format threshold keep stock bits, so a verify row equals the same row decoded alone only in `exact` mode (`min_rows=1`). The lane law id joins the APCv2 namespace, so state computed under one law is never reused under another. KR-04 (a grouped member computed with a different split-K above 32 rows) is fixed: wide calls now chunk the group's stacked launch (016fe802e). On the 27B ordinary route the lane kernel barely engages (decode rows are below the 8-row threshold), which is why the lane-off arm in chapter 11 differs only by noise.

Every row threshold in one place

MECHANISM	ROWS WHERE BEHAVIOUR CHANGES	SOURCE
Lane matmul crossover	quantized ≥ 8 rows, bf16/fp16 ≥ 16 rows; stock above 32 (128 only with external varlen + a tree route)	<code>lane/policy.py</code>
Tensor-unit fragment	16 rows per fragment; 17 rows take two	<code>lane/matmul.py</code> (<code>ROW_BLOCK</code>)

MECHANISM	ROWS WHERE BEHAVIOUR CHANGES	SOURCE
Tree GDN state slots	16-slot kernel up to 16 rows, then 32; chains up to 128	<code>qwen38_tree_gdn.py</code>
Fused GDN verify (Flash-Next)	default ≤ 8 steps; bit-exact proven to 17; wider refused	<code>qwen4_fused_gdn_verify.py</code>
Self-MTP verify rows	Qwen3.8 declares 9 (8-token copy + anchor); Flash-Next 17	adapters (exact row capability)
Tree budget (Qwen3.8 + DFlash2)	about 16 target rows per round at B1–B8; chains shrink to 0 drafts from B9	<code>external_speculative.py</code>
MoE NAX gather admission	≥ 16 rows and ≥ 4 rows per expert, prefill only; affine 4/6/8-bit + MXFP4; sorted route from 20 assignments	<code>moe_nax_gather.py</code> , <code>switch_layers.py</code>
Floor-3 rhs pad	sorted rows in [3E, 4E) padded to $\max(4E, 16)$	<code>switch_layers.py</code>
QSA stage-one kernel	query ≥ 64 and physical KV $\geq 65,024$ (NAX prefill from 8,192 KV)	<code>qwen4_exp.py</code> , <code>flash_next_policy.py</code>
Prefill autoscale (no adapter step)	prompt $\leq 32K \rightarrow 512$ -row slices, $\leq 64K \rightarrow 2048$, $\leq 128K \rightarrow 8192$	<code>prefill_plan.py</code>



WHAT THE 2026-10-06 MEASUREMENTS FOUND (Qwen3.6-35B-A3B Heretic 4-bit, 8K prefill, M5 Max)

1.077×

NAX fused vs off, end to end
1,565 vs 1,685 ms per 8K prefill

1.87

median CV of rows per expert;
busiest expert 951 rows vs mean 64

0.5–0.7%

of prefill recoverable by a
per-call best plan: don't build it

1.071×

6-bit NAX gather, 35B 8K prefill
1,513 $\rightarrow$ 1,413 ms, bit-identical

Admission mirrors MLX (≥ 16 rows, ≥ 4 per expert), prefill only; affine 4/6/8-bit + MXFP4 by default (6-bit since 7e28b1ace; 2/3/5 exact, not admitted). Same $16 \times 32 \times 16$ ops in the same K order as MLX $\rightarrow$ bit-identical; every kernel instantiation is canaried against the stock path. The floor-3 rhs pad (default, unqualified, now named in receipts) pads 3E–4E sorted rows to 4E and switches MLX from `gather_qmv` to `gather_qmm_rhs`.

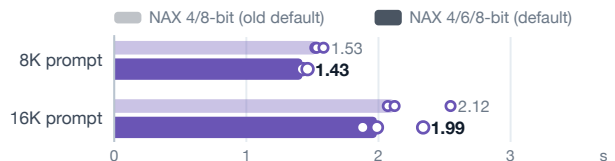
Figure 6.2. MoE NAX sorted gather. Top: the mechanism, schematic. Bottom: the 2026-10-06 research measurements (qualification/runs/research-20261006/b1-moe-tile-headroom, mlx2 f30cce87d, and nax-6bit). The artifact turned out to have 6-bit experts in 20 of 40 layers, which NAX did not serve (stock gathers took 21.5% of prefill); that finding, not a tile planner, became the next kernel.

6-bit experts on the NAX path

46656a8ec teaches the NAX sorted gather to unpack affine 2/3/5/6-bit weights and routes admission by width; 7e28b1ace admits 6-bit by default (`MLX2_MOE_NAX_GATHER_BITS=4,8` restores 4/8). 102 canary variants of the 6-bit kernel are exact against the stock path, and at the real 35B shapes logits and 24 greedy tokens are bit-identical. 2/3/5-bit are exact in canaries but not admitted. Served end to end, the default moves median 8K TTFT -7% , 16K TTFT -6% and B8 aggregate $+7\%$ (ledger, by rep ranges: -7% , -8% , $+4\%$); B1 decode does not change (155 vs 154 tok/s) because NAX runs only in prefill. It is a **new default**: routes that serve 6-bit experts need requalification.

Cold-prompt TTFT

Qwen3.6-35B-A3B Heretic, ordinary, 3 reps, s · lower is better



Evidence: qualification/runs/sweep-20261006/final-q36-6bit · main 43d561e38

Flash-Next

Gated DeltaNet, QSA sparse attention and a 512-expert MoE in one 48-layer model.

Flash-Next (the Qwen4-Exp architecture, served here as Qwen3.8-Flash-Next-Uncensored 4-bit with an MTP head) was mlx2's first model and still carries its densest set of selected mechanisms. It is a hybrid in every dimension: recurrent and attention layers, sparse and dense attention, routed and shared experts.

48 LAYERS · hidden 2560 · every layer has a routed MoE



PLE

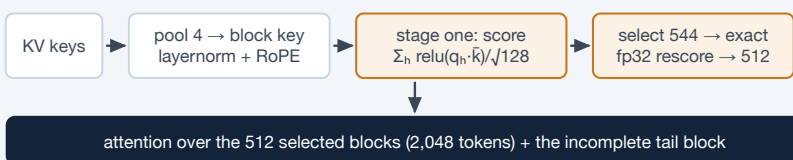
teal = Gated DeltaNet (36) · amber = full attention + QSA indexer (12; layers 3, 7, ..., 47) · hyper-connections × 4 · MTP head on top

ROUTED MoE (each layer)

router: softmax → top-10 of 512 experts, renormalized; fused Metal router at width 1 expert MLP 640 wide + shared expert with sigmoid gate

prefill: sorted gather on NAX (fused split SwiGLU), adaptive rhs pad
decode: routed top-k fold, launch path

QSA SPARSE ATTENTION (full-attention layers)



Stage-one kernel engages at query ≥ 64 and physical KV ≥ 65,024; otherwise stock argpartition selects. Default producer/selector mpp_exact_band + radix_exact; direct4 is opt-in (env only).

FUSED GDN (36 layers): one kernel family per phase, bit-identical to the reference by construction or proof

decode

packed recurrence:
8 value rows / SIMD-group

verify

≤ 8 steps default,
proven exact to 17

rollback

compact replay tape:
keys, corrections, decay

prefill

fused prefill kernel
(chunk16 research: inexact)

batch

row_exact: B=1 kernel
per lane, identical bits

Fallbacks are counted with a reason (uninitialized cache, no rollback records, width above max steps, masked verify ...).

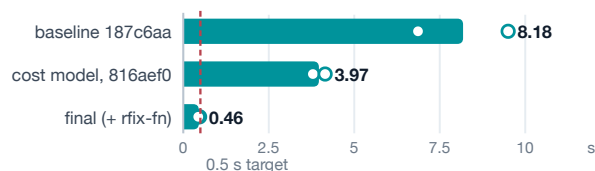
Figure 7. Flash-Next anatomy and its three signature mechanisms. Layer layout from the artifact config (full attention every 4th layer); QSA geometry from `qwen4_exp.py` (compress ratio 4, indexer budget 2048 → 512 blocks, 4 heads × 128) and `qwen4_qsa_stage1.py`; fused GDN phases and limits from `flash_next.py`, `flash_next_policy.py` and `qwen4_fused_gdn_verify.py`. MTP: K = 2, ordinary handoff above cohort width 3, copy drafts on, interior checkpoints auto.

The decode stall, fixed in three steps

The baseline campaign's worst result was here: with the served policy, a decoding user beside a 16K prompt saw a token gap of 6.9–9.5 s (up to 9.9 s with QSA `direct4`; ordinary decode, on the same model, about 0.5 s). On self-MTP a residual that fit the uncontended 8,192-row step was prepared whole beside live decode. The cost model (d28a774aa) bounded that preparation, and the intermediate rerun cut the maximum to 3.8–4.1 s, but it also exposed a new defect: eight co-arriving prompts were prepared in waves (TTFTs 1.25 / 8.9 / 12.2 s) because the first idle boundary had admitted only two of them. The review-fix round then made three changes: co-arrivals left over from a partial first boundary are prepared together (057373692, figure 4.3), the hold is bounded per sibling (e4213207f), and per-layer-embedding (PLE) prefetch stages one slice ahead instead of re-staging the whole tail on every slice (it had staged 3.0 M rows for 0.24 M used on one 16K prompt, host work invisible to the stall bound; daece1c6d).

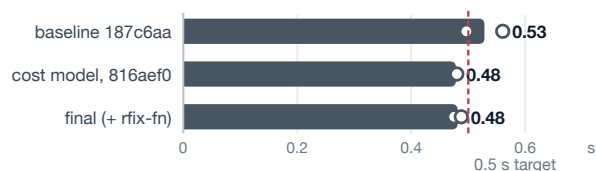
Served MTP: max decode gap beside a 16K prefill

Flash-Next, s, per stage of the fix · lower is better



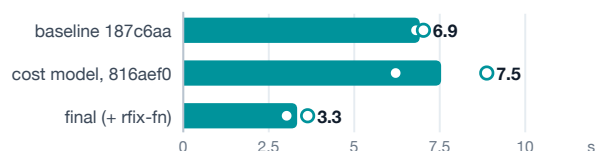
Ordinary decode: max gap, same workload

Flash-Next ordinary route, s, per stage · lower is better



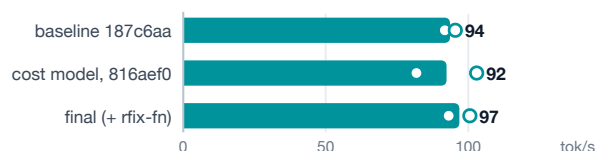
Served MTP B8: median TTFT

8 co-arriving prompts, s · lower is better



Served MTP B8: batch throughput

tokens ÷ (first submission → last token), tok/s · higher is better



Bars are medians, rings repetitions. Final: max gap 440–475 ms (ordinary 475–488 ms). B8 is measured as batch wall time: an earlier first-token-to-last-token aggregate (143–159 → 87 → 112–126 tok/s) rewarded staggered starts and is retracted in the GPU ledger. On wall time the batch finishes in the same time as before and the median user gets a first token about twice as fast. Evidence: [qualification/runs/sweep-20261006/fn](#) · Evidence: [qualification/runs/sweep-20261006/post-fn](#) · Evidence: [qualification/runs/sweep-20261006/final-fn](#)

QSA SELECTORS

QSA `direct4` was neutral in the baseline (B1 84.5 tok/s; 65.3 vs 59.0 tok/s decode after a 32K prompt); it stays `env-only` and `default-off`, with a truthful status (3e82bb6c9). A GVR top-k selector (223068567, `research-20261006/b2-qa-gvr`) is exact and, on a synthetic grid, runs the selector at a median 0.75× of `direct8`'s time and the complete stage one at 0.83×; heavy-tie inputs fall back and are slower than both. It is **OPT-IN**, unqualified, and has not run end to end.

OTHER FIXES THAT TOUCH FLASH-NEXT

- QSA exact-band refine excluded valid zero-score blocks behind `-inf` sentinels a `relu` clamped (KR-01, dcb7c25df; latent below the 65,024-KV floor).
- NAX QSA prefill declines masks with extra structure (KR-09, 0bc1ad162) and now requires registered plain-mask provenance (3ce12a531).
- The research-only `fused_outproj` GDN decode kernel (grid-wide spin barrier on relaxed atomics) is refused (KR-08, 94dd01820).
- COMPACTION is no longer declared (9fcfc4acb); fused GDN prefill is a required observation (734a79679).

Memory admission

Reserve + prefill transient + lane cost must fit measured headroom before anything is allocated.

On Apple silicon the GPU has no memory of its own: weights, KV and recurrent state, the prefix cache, transient activations, the OS and every other app share one pool. Running out is not an exception mlx2 can catch; it is swap, a GPU timeout or a killed process. So admission is decided before any allocation, from measured headroom.

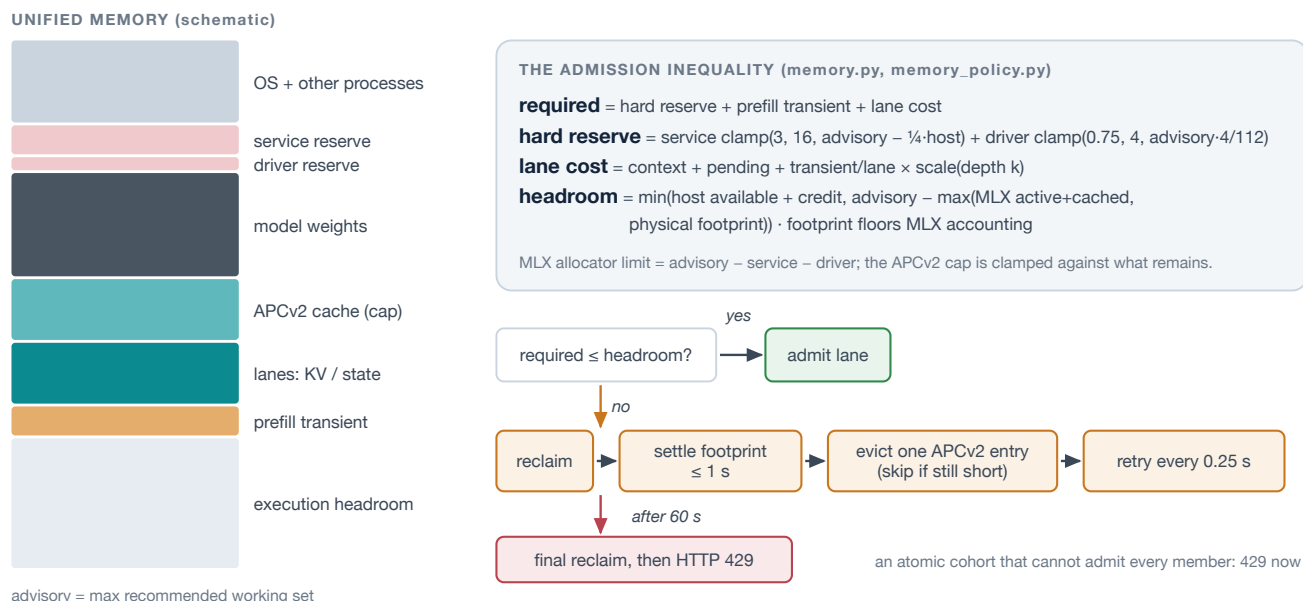
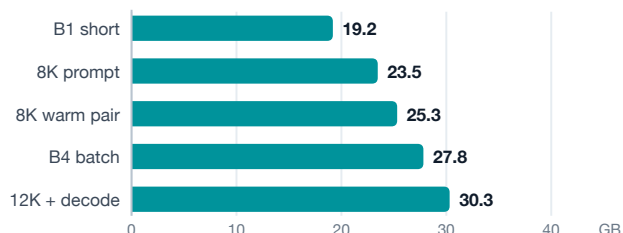


Figure 8. Memory admission. Left: the pool, schematic and not to scale. Right: the inequality evaluated before each lane is admitted, and what happens when it fails. Constants: `CACHE_GIB_PER_1K_TOKENS = 0.44`, transient scale by depth `{0: 1/3, 1: 0.8, 2: 1.0, 3: 1.25, 4: 1.55}`, footprint settling tolerance 512 MiB within 1 s, retry every 0.25 s up to 60 s. Sources: `memory.py`, `runtime/memory_policy.py`, `serving.py` (`ensure_admission_headroom`, `lane_admission_required_gib`).

Peak Metal memory by workload

Qwen3.8-27B ordinary, served arm, median of 2 reps · GB ·

`qualification/runs/sweep-20261006/q38-features`



WHY THE FOOTPRINT, NOT THE ALLOCATOR

MLX's accounting (active + cached) can lag what the process holds while Metal releases pages, so admission floors it with the live `physical_footprint` and waits up to 1 s for freed pages to settle. Grants not yet materialized are subtracted, so two admissions cannot spend the same headroom.

ONE HEADROOM FUNCTION

The opt-in host floor (`minimum_host_available_gib`) capped lane admission but not the parallel-sample ($n > 1$) fan-out guard, contradicting `docs/SERVING.md`; both now use one headroom function (`fc22e2885`).

Power, thermal and the governor

Joules, DVFS residency and die temperature, read without privilege, attributed to requests and, opt-in, used to hold a power cap.

`--power-telemetry (OPT-IN)` publishes what the server draws, without root. The same sources power the profilers that produced this document's J/token figures and `mlx2-top`, which reads IOReport by default. New since the first round: a governor that acts on those readings, also opt-in, and a review round that hardened the native samplers (results checked and released, sampling serialized, only complete GPU + DRAM readings fed to the governor, attribution window bounded by seconds: f52578e61, f03a51b6e, dffad06da, 01fd7a3a2).

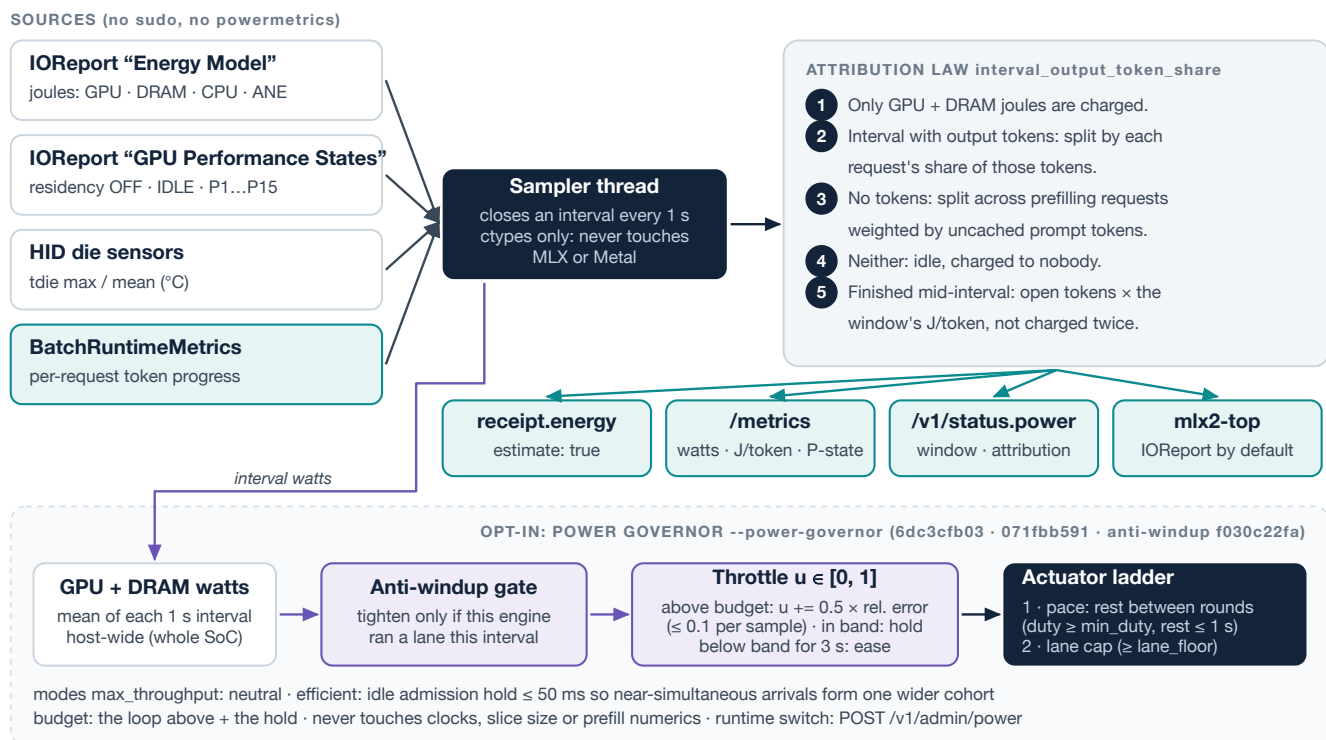
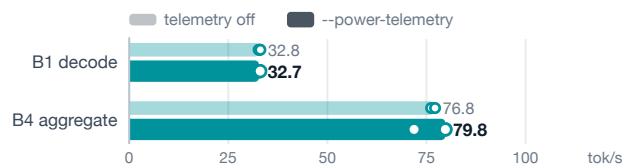


Figure 9. Telemetry pipeline and the governor. All sources are readable by an ordinary user; the sampler binds CoreFoundation, IOKit and libIOReport through ctypes and never touches MLX or Metal, so it cannot stall or reorder device work. Joules are whole-SoC readings: concurrent requests are not separable in hardware, so per-request energy is an allocation rule (receipt `mlx2.energy-estimate.v1`, `estimate: true`). The governor (bottom) never touches clocks: macOS gives userspace no DVFS control. It paces rounds first and caps lanes last, because the thermal map shows pacing is the only lever that lowers watts. Sources: `docs/SERVING.md` (Power governor), `power_governor.py`, `power_telemetry.py`.

Telemetry overhead

Qwen3.8-27B ordinary, 3 reps, tok/s ·

`qualification/runs/sweep-20261006/telemetry-overhead`



The sampler costs about 6 ms (IOReport) + 30 ms (HID, GIL released) per 1 s interval on a background thread. Its effect on decode is within run-to-run noise at B1 and B4 (integration tree 816aef02c; identical code, flag on vs off), so the profilers can keep it on.

The thermal map: a burst, then a budget

The thermal map loads ordinary, native MTP and DFlash2 servers for Qwen3.8-27B at once and drives them in strict alternation through 8K prefills, B1 and B4 decodes, so every route is compared at the same heat: a cold start, 25 minutes back to back, a cooldown, and 11 minutes with 3 s gaps (`scripts/thermal_map.py`). The machine behaves like a power-budgeted part, not a thermally throttled one.

89 W

Cold burst, GPU, first prefills

892 tok/s for about 90 s

41 W

Sustained budget, GPU, late prefills

held for the 25-minute hot phase

63–66 °C

Die plateau (hot steady)

power-limited, not temperature-limited

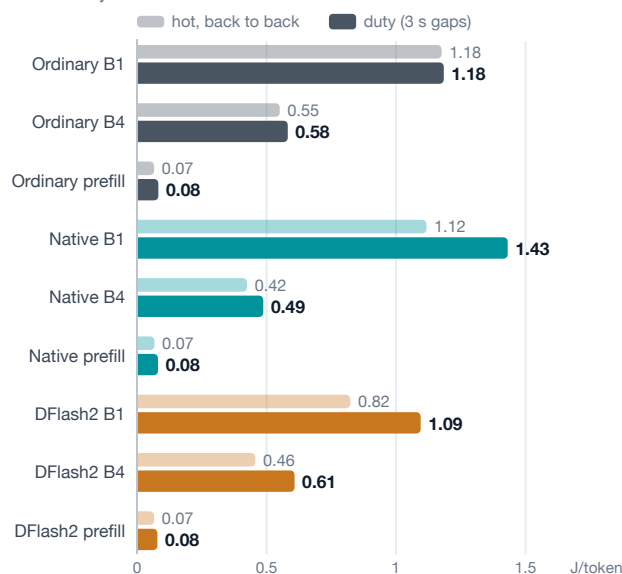
+17–24 %

J/token with idle gaps

pacing is not an efficiency lever

Energy per token: steady hot vs 3 s idle gaps

GPU + DRAM joules ÷ tokens · lower is better

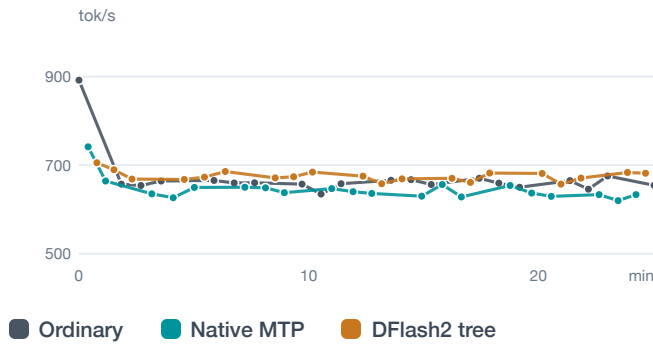


What the map says, per the GPU ledger: B4 decode is 2.1–2.6× more energy-efficient per token than B1, and DFlash2 B1 spends 0.82 J/token against ordinary's 1.18. Ordinary B1 decode is memory-bound, and DVFS parks it at P6–7 by itself. With 3 s gaps each segment runs faster (prefill +10%, about 55 W) but every route spends 17–24% more energy per token, and wall-clock decode falls from 30 to 23 tok/s. So the burst is a latency resource to spend on interactive TTFT, and the efficient levers are wider cohorts and the better route, not rests. Prefill is the phase that loses the most when hot (–26% from the burst); B1 decode loses about 6%.

Evidence: `qualification/runs/sweep-20261006/thermal-q38 (summary.json, segments.jsonl) · origin/main 187c6aa23`

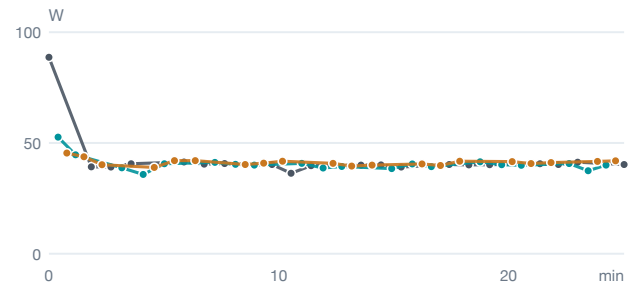
8K prefill rate through the hot phase

each point one prefill segment, tok/s



GPU power during those prefills

IOReport GPU watts per segment

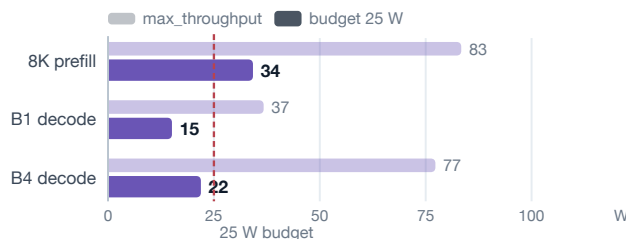


The governor on the GPU

The governor (figure 9) is implemented and **OPT-IN**. Two campaigns ran it on Qwen3.8-27B ordinary with power telemetry on, several servers loaded and strict rotation. The final run (anti-windup build) compares **max_throughput** with a 25 W GPU + DRAM budget for 15 minutes: the budget held the average at **24.5 W** against 59.7 W uncapped, for decode 17.9 vs 54.0 tok/s and prefill 320 vs 800 tok/s, at 0.181 vs 0.158 J/token (ledger averages while active). About –59% power costs about –67% throughput and +15% energy per token.

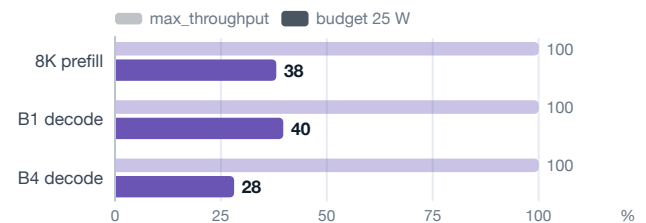
GPU + DRAM power

hot segments, W · the cap applies to this sum



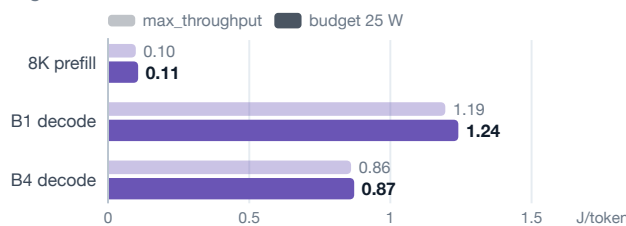
Throughput kept under the cap

budget rate ÷ max_throughput rate, % · higher is better



Energy per token

hot segments, J/token · lower is better



Pale = **max_throughput**, solid = budget 25 W, per segment of the strict rotation (hot phase medians). The cap is enforced by pacing: one duty cycle for the whole rotation, so every segment keeps a similar share of its rate (28–40%), and B1 decode drops to about 15 W although it alone draws about 38 W uncapped. Single segments sit above or below 25 W; the controller holds the 30 s rolling mean.

Per-segment hot medians from [qualification/runs/sweep-20261006/final-governor/summary.json](#) · main 43d561e38.

The first run (below) found an apparent windup: an idle budget-mode server tightened on another server's power and paced its own B1 decode to 11 tok/s. Anti-windup (tighten only in intervals this engine ran a lane, f030c22fa) was added, but the rerun showed the B1 throttling was correct: B1 decode draws about 38 W GPU + DRAM (about 20 W GPU only), so a 25 W GPU + DRAM budget must pace it. Anti-windup stays, for hosts shared with other work.

ALL FOUR MODES (INTEGRATION TREE 816AEF02C, 20 MINUTES, HOT MEDIANS)

MODE	8K PREFILL			B1 DECODE			B4 DECODE		
	TOK/S	W	J/TOK	TOK/S	W	J/TOK	TOK/S	W	J/TOK
max_throughput	746	59	0.08	32	39	1.22	82	68	0.82
efficient	746	60	0.09	31	38	1.23	87	70	0.81
budget 25 W	322	34	0.10	11	15	1.30	23	22	0.94
budget 15 W	318	34	0.11	8	11	1.36	15	12	0.82

`efficient` holds the idle admission window up to 50 ms to form wider cohorts. On this workload (requests arrive one segment at a time) its whole-run energy per token was only about 2% below `max_throughput` (ledger: 0.162 vs 0.165 J/token), less than the per-segment medians above can resolve. Budget mode is a reliable hard cap and a grid demand-response lever, not an efficiency mode.

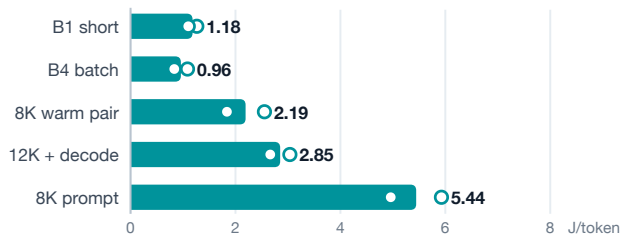
Evidence: [qualification/runs/sweep-20261006/governor](#)

Energy per token in the baseline campaigns

Energy per output token, Qwen3.8-27B

served arm, GPU + DRAM joules ÷ output tokens · lower is better ·

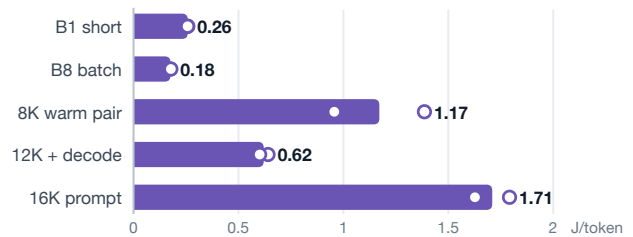
[qualification/runs/sweep-20261006/q38-features](#)



Energy per output token, Qwen3.6-35B-A3B

ordinary (no MTP head in this artifact) · lower is better ·

[qualification/runs/sweep-20261006/q36](#)

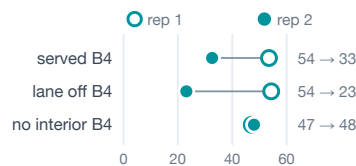


DVFS is visible, and it matters for comparisons

The same configuration, rerun later in the campaign, can sit at a different GPU performance state. For the B4 batch below, the second repetition of the served arm drew about 39% less GPU power for about 9% less aggregate throughput, so it spent less energy per token. The lane-off arm's second repetition dropped further (mean P-state 11.2 → 6.1), which is why its chapter 11 median looks 8% slower: that is the clock, not the kernel. Rep order was rotated; die temperature was not higher in rep 2, so the cause is the governor's choice, not throttling.

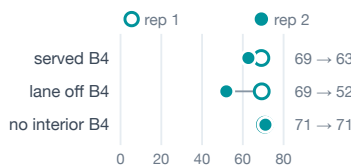
GPU power

W



Aggregate decode

tok/s



Energy per token

J/token



Hollow = repetition 1, filled = repetition 2, same arm and workload. [qualification/runs/sweep-20261006/q38-features/results.json](#) (energy blocks per workload).

Qualification

Implemented, selected, observed-used and qualified are four different facts with four kinds of evidence.

Qualification is a statement of confidence, not permission to run: evidence that a route loads and decodes properly, that its defaults work and that nothing regressed badly. An unqualified route still serves and says so in `/v1/status` and every receipt. It is not performance testing.

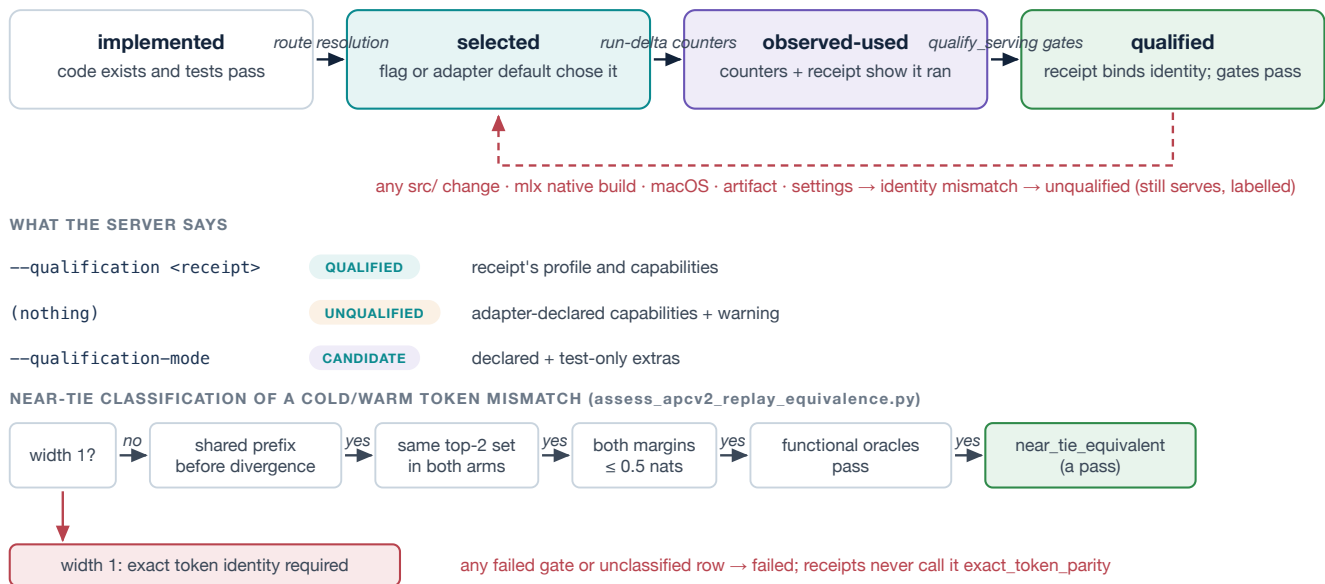


Figure 10. Qualification states and the near-tie contract. Top: the four states and the evidence that moves a route between them; the rose edge is invalidation by identity. Middle: the three server modes. Bottom: how a cold/warm token mismatch at physical width above one is classified (0.5-nat ceiling; AGENTS.md, QUALIFICATION.md, `scripts/assess_apcv2_replay_equivalence.py`). The verdict always carries `qualification_claim: false`; a classified near tie is a pass, never exact parity.

A REAL RECEIPT (DFLASH2, B1)

```
route: external_draft
qualification: unqualified
speculation.accepted / proposed: 97 / 510
batch_size_route.policy: tree15_b1_b4_chain_b5plus_v1
tree_node_budget_by_lanes: {'1': 15, '2': 7, '3': 4, '4': 3}
batch_size_route.qualified: False
```

From `qualification/runs/sweep-20261006/final-q38-routes/results.json`.

WHAT THE SWEEP CHANGED

- **Selected ≠ observed-used can no longer qualify** (CQ-4, 734a79679): default-on mechanisms are now required as run deltas.
- **The default Qwen3.8 tree route can qualify** on observed tree rounds (93a8e99de); every other tree route stays a candidate.
- **Defaults that can never fire step aside** (36505df39); `MLX2_*` switches (e385f6be5) and the fairness estimator (acadd8b76) are in route identity; external-draft features are run deltas (015c44843).

Route comparison

Every route through the serving path: origin/main 187c6aa23 before the sweep, and the final tree after all fixes.

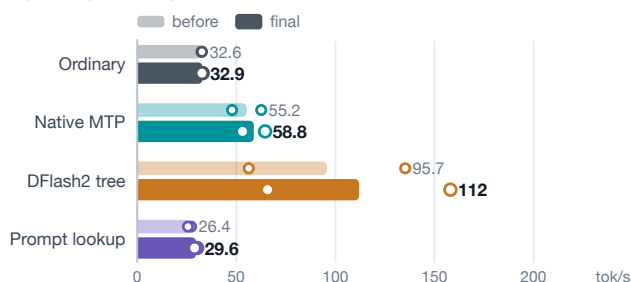
All measurements in this chapter were taken through the real HTTP serving path by `scripts/profile_features.py`, one arm (server configuration) per process, on an M5 Max with 128 GB. **Before** is origin/main 187c6aa23 plus default-inert fixes; **final** is main 43d561e38 with every fix of the sweep, the review-fix round and the integration fixes. Both used the same specs, workloads and repetition counts. Every route was unqualified at both sources. Speculative routes are exact in law, but at temperature 0 their text can still differ from ordinary decode at numerical near ties; the profiler's identity column compares output hashes, it is not the near-tie classifier.

Qwen3.8-27B: four routes, before and after

■ Ordinary ■ Native MTP ■ DFlash2 tree ■ Prompt lookup

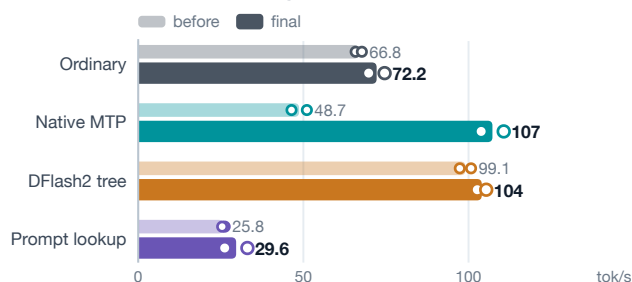
B1 decode, short prompt

tok/s per request · higher is better



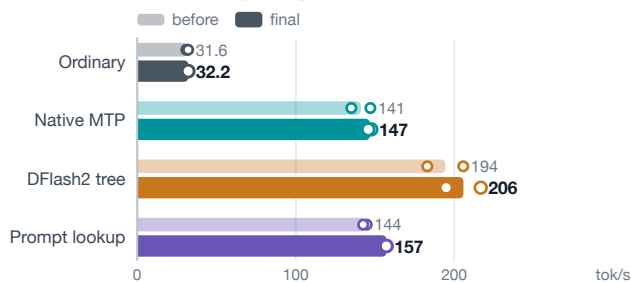
B4 aggregate decode

4 concurrent 700-token notes · higher is better



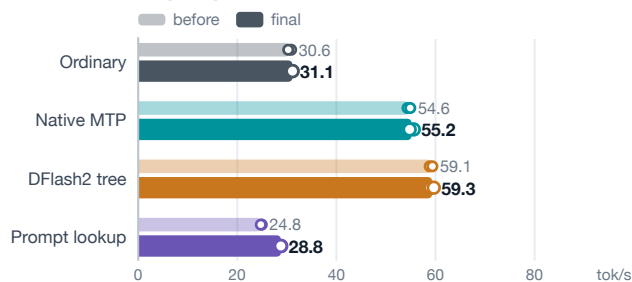
Copy workload

reproduce a 450-token excerpt · higher is better



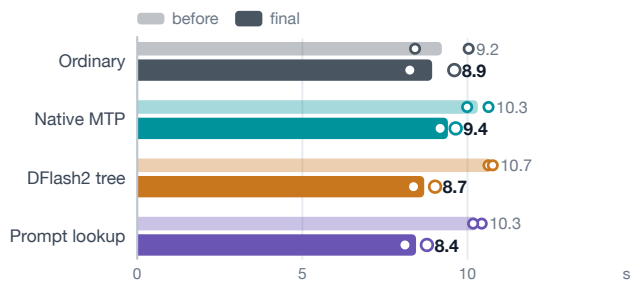
JSON-schema output

constrained decoding · higher is better



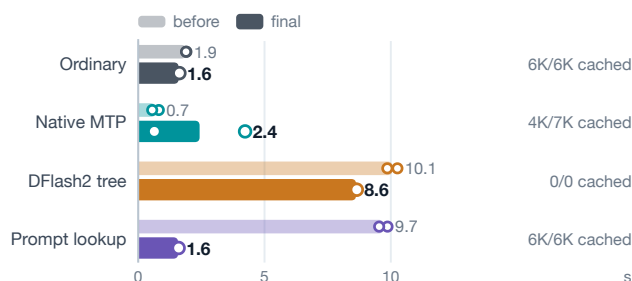
Cold 8K prompt, TTFT

first token after a fresh 8K document · lower is better



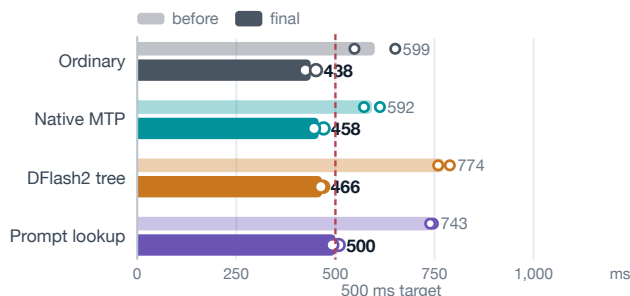
Warm 8K pair, TTFT

same document, new question · final cached tokens per rep · lower is better



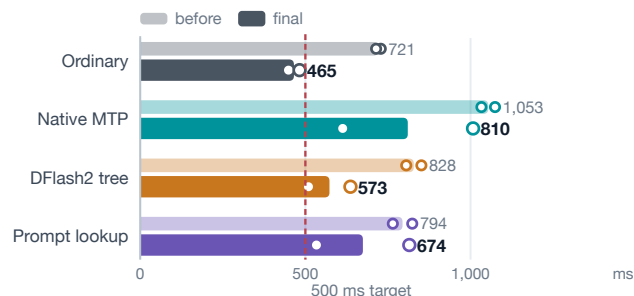
Neighbour gap p95 beside a 12K prefill

decoding lane's token gaps while the prompt prefills · lower is better



Neighbour gap max, same window

outliers remain on MTP and PLD · lower is better



Qwen3.8-27B oQ4e-mtp · `--max-lanes 4 --max-context 32768 --cache-bytes 8589934592 --lane-matmul auto --default-max-tokens 4096` · 2 reps rotated, 30 s cooldown, max_tokens 256, swapouts Δ 0 before / 0 final. Bars are medians, rings are repetitions; pale = before, solid = final.

Tool-call TTFT, final: Ordinary 1.47 s · Native MTP 0.91 s · DFlash2 tree 0.59 s · Prompt lookup 1.04 s (before: 1.30 · 0.98 · 0.63 · 1.21 s). Final output identical to ordinary (hash): **short** Native MTP: 1/2 · DFlash2 tree: 1/2 · Prompt lookup: 1/2 · **batch:4** Native MTP: 2/8 · DFlash2 tree: 1/8 · Prompt lookup: 0/8 · **copy** Native MTP: 2/2 · DFlash2 tree: 2/2 · Prompt lookup: 2/2 · **json** Native MTP: 2/2 · DFlash2 tree: 2/2 · Prompt lookup: 2/2. Evidence: [qualification/runs/sweep-20261006/q38-routes](#) · Evidence: [qualification/runs/sweep-20261006/final-q38-routes](#)

READING THE 27B PANELS

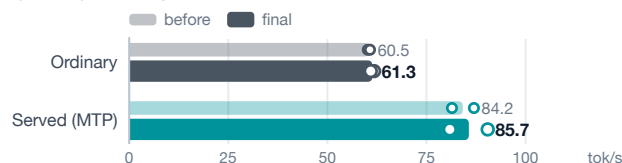
- **Native MTP B4 more than doubled** (cohort fix SS-1) and is now 1.5× ordinary; DFlash2 B4 rose about 5% and ordinary B4 about 8%.
- **Every route now holds the p95 neighbour gap at about 500 ms or under**, including DFlash2 and prompt lookup, which the first round did not bound (673aa9fe0). Maxima still exceed the target on MTP (up to about 1 s), PLD and DFlash2: outliers, now attributable (d0ac480ae), not yet removed.
- **Prompt lookup warm reuse is fixed** (recurrent checkpoints, cb26c206b): its warm TTFT now matches ordinary. **Its B4 is not**: the cohort is batched (246 rounds, average width 3.5), but 814 of 855 cycles are plain-decode rounds at about 114 ms each, about 2× an ordinary B4 step, because they go through the hybrid segmented verify path. A fix is in progress (open).
- **Native MTP warm hit depth is unstable**: one rep reused 4,096 tokens (4.2 s), the other 7,168 (0.64 s). **DFlash2 warm reuse stays absent by design** (no paired draft state below the prompt boundary).
- The cold 8K TTFT differences are heat, not route: the baseline's 8.4 s ordinary first run was the cold machine (ledger correction).

Flash-Next: served policy versus ordinary, before and after

■ Ordinary ■ Served (MTP)

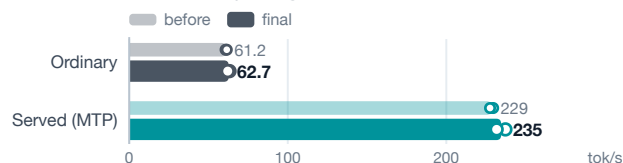
B1 decode, short prompt

tok/s per request · higher is better



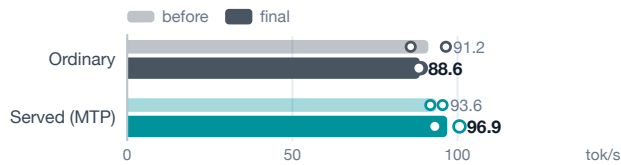
Copy workload

reproduce a 450-token excerpt · higher is better



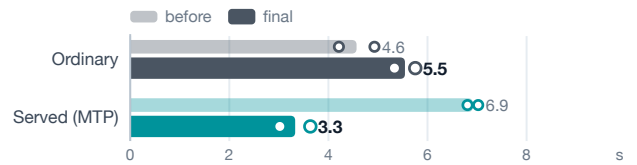
B8 batch throughput

tokens ÷ batch wall time · higher is better



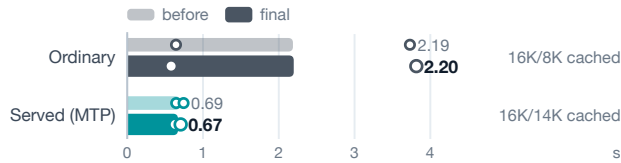
B8 median TTFT

median of the 8 requests · lower is better



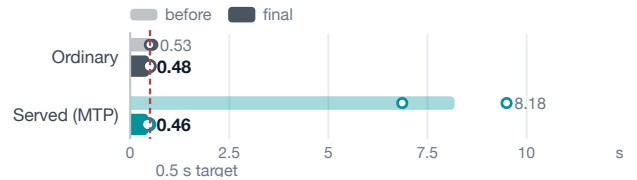
Warm 16K pair, TTFT

final cached tokens per rep at right · lower is better



Decode stall beside a 16K prefill

max neighbour token gap · lower is better



Flash-Next Uncensored 4-bit MTP · `--max-lanes 8 --max-context 65536 --cache-bytes 12884901888 --default-max-tokens 4096` · served = native MTP K=2, handoff above width 3. The baseline also ran QSA `direct4` and a 32K prompt (chapter 7); the final campaign did not. Evidence: [qualification/runs/sweep-20261006/fn](#) · Evidence: [qualification/runs/sweep-20261006/final-fn](#)

READING THE FLASH-NEXT PANELS

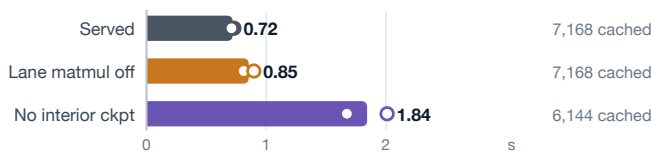
The stall beside a 16K prefill is fixed on the served route (6.9–9.5 s → under 0.5 s; chapter 7 shows the three steps). B8 is measured as batch wall time: the batch takes as long as before, and the median request's first token arrives about twice as early.

Qwen3.8-27B: interior checkpoints and lane matmul (baseline)

■ Served ■ Lane matmul off ■ No interior ckpt

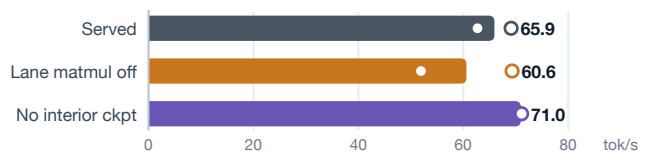
Warm 8K pair, TTFT

interior checkpoints on vs off · lower is better



B4 aggregate decode

lane matmul auto vs off · higher is better



Served arm = lane matmul auto + interior checkpoints auto; the lane-off B4 median is pulled down by DVFS (chapter 9). Rerun with `--qualification-mode (q38-features2, ledger only)`: int8 MLP prefill is refused by the 27B adapter (no int8 scope; fail-closed, correct) and GDN chunk 16 stays rejected (noise end to end, inexact). Evidence: [qualification/runs/sweep-20261006/q38-features](#)

Qwen3.6-35B-A3B Heretic: ordinary

136 tok/s

B1 decode

0.26 J/token

251 tok/s

B8 aggregate

0.18 J/token

4.3 s

Cold 16K TTFT

final with 6-bit NAX 2.0 s

196 ms

Max decode stall beside 12K

well inside the 500 ms target

No MTP head in this artifact: ordinary route only; the final 35B campaign measured 6-bit NAX (chapter 6). Evidence: [qualification/runs/sweep-20261006/q36](#)

What the sweep fixed

Every finding, its severity, the commit that addressed it, the GPU check and what is true now.

156 non-merge commits on top of origin/main 187c6aa23 fixes, receipts, docs, research, telemetry	62 ledger rows fixed 3 P0 · 28 P1 rows fixed or partial	18 rows with a GPU check named in the GPU column	7 partial or design-only kept visible, not claimed	5 still open see chapter 14
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Each row is a finding from a sweep lane report (ids as in `qualification/runs/sweep-20261006/lanes/*.md`), the GPU ledger, the review-fix round, the Codex review or integration, with the commit(s) that addressed it and what is true now. **Fixed** means changed and covered by a test or repro on CPU; the **GPU** column names the final (or identical-code) A/B that confirms the effect, where one exists. **Partial** means the reported defect is addressed but a named remainder is not (for example a fix that batches a route but leaves a named cost for a later commit). Severity: P0 breaks a default route; P1 wrong output, wrong evidence or a large loss on a default path; P2 latent, perf or hygiene.

SCHEDULING AND SERVING

ID	SEV	FINDING	COMMIT	GPU	NOW
SS-1	P1	Co-arriving multi-chunk prompts locked the self-MTP cohort at width 1; siblings decoded plain beside it	56fce148e e4213207f	MTP B4 46–51 → 104–111	FIXED
SS-2	P1	Stall bound used a running-max, row-blind rate pooled across kinds (Flash-Next 6.9–9.9 s stall)	d28a774aa	A/B p95 –16..–23%	FIXED
SS-3	P1	Self-MTP burst preparation unbounded and invisible to fairness	d28a774aa 470e7d3af	FN stall → 0.44–0.47 s	FIXED
SS-4	P1	Autoscale sized one warm request from two different lengths	2d4a9a676		FIXED
SS-5/6	P2	batch_geometry inert beside SRPT; no-op knob in route identity	4a0c00bae		FIXED
SS-7	P2	Row tile hard-coded in three places, not adapter-derived	48289d962 d28a774aa		FIXED
SS-8/10	P2	Clamp counters undone by the slice floor; MTP deadline path diverged	d28a774aa		FIXED
SS-9	P2	Native continuations invisible to the ordinary scheduler (default-off routes)	–		OPEN
API	P2	Raw completions lost the first token's leading space; lone surrogates gave 500; tool-search blocks refused	fb8b939f5 70a989ab2 fc7ca8777		FIXED
new	–	GPU keep-warm ticker for idle gaps	726837c12	–6% TTFT at 20 s idle	OPT-IN

ID	SEV	FINDING	COMMIT	GPU	NOW
new	—	Unprivileged power and thermal telemetry, published by the server	a809a970d b2367195d	overhead in noise	OPT-IN
new	—	Power governor: efficient hold and closed-loop budget, admin endpoint	6dc3cfb03 071fbb591	cap held; ~67% tok/s	OPT-IN

APCV2 AND STATE

ID	SEV	FINDING	COMMIT	GPU	NOW
APC-1	P1	A deeper sidecar-less entry hid a usable MTP sidecar → full miss on self-MTP	f6a160933 1ee34dd68		FIXED
APC-3	P1	Warm continuations cut prefill at different positions from cold	d23b16895		FIXED
COW-1	P1	Shared checkpoint snapshots charged once per entry (up to ~5×)	cc4e96bac b02360126		FIXED
APC-2	P2	Hit counters per lookup, not per request; refused hits kept credit	cd9fb13b3 73548bdba		FIXED
COW-2/3	P2	Rewritten borrowed recovery accepted; approximate staging mutated its source	ea3f41677		FIXED
PAGED	P2	Paged index/bridge/private cache hardening (no serving caller)	b02982540		PARTIAL
GPU	P1	Prompt lookup recorded no recurrent checkpoints → warm 8K TTFT 9.7 s vs 1.9 s	cb26c206b	PLD warm 9.5–9.9 → 1.6 s	FIXED
CQ-13	P2	MLX2_* switches changed the law without changing settings or the APCv2 key	e385f6be5		FIXED
design	—	Value-aware retention (saved prefill per unique byte)	f44f355b5		OPT-IN

SPECULATIVE DECODING

ID	SEV	FINDING	COMMIT	GPU	NOW
SPEC-10	P1	Prompt lookup never batched on hybrid targets (B4 = 25.8 vs 66.8 tok/s ordinary)	2942ee30b 1c45a38a7 82bdcc230	B4 25–26 → 55–63 (ordinary 62–63)	FIXED
SPEC-01	P1	pipeline_draft built and discarded a lattice per lane per round at B2–B4	fb21b8a41		FIXED
SPEC-03	P1	B5+ chain leg jumped to 8 rows per lane, unmeasured	1f9c22d1d		FIXED
SPEC-04	P1	Composition rebuilt a full-history index per row per round (161 ms at 128K)	6f6fe1118		PARTIAL
SPEC-02	P1	Near-tie contract cannot fail on rate or cross-arm shift	—		OPEN
SPEC-05/07	P2	Block verify untested; zero residual mass raised instead of drawing from p	34216acbb		FIXED
SPEC-06	P2	Greedy chain verify did dense O(K·V) host work (~3.6 ms per lane-round)	883c0b638		FIXED
SPEC-08/09	P2	Stale default-off labels; copy refusal reported as lookup miss	4913b60ab 9d53f14bc		FIXED
CQ-11	P2	The new Qwen3.8 tree default could never qualify	93a8e99de		FIXED
new	—	Exact block verification for the linear external chain	ccbf2a99c	not run	OPT-IN

KERNELS AND ROWS

ID	SEV	FINDING	COMMIT	GPU	NOW
KR-05	P1	Floor-3 sorted-MoE rhs pad default-on, unqualified, missing from receipts	496288096 6cf03583e		PARTIAL
KR-01	P2	QSA exact-band refine dropped valid zero-score blocks (latent)	dc7c25df		FIXED
KR-02	P2	QSA direct-selector status lied; unknown values became off	3e82bb6c9		FIXED
KR-03	P2	Varlen compaction declines were silent; host loop per row	6b5f6aed3		FIXED
KR-04	P2	Lane-matmul row bits depended on call width above 32 rows	016fe802e		FIXED
KR-08/09	P2	Spin-barrier GDN kernel; NAX QSA ignored extra mask structure	94dd01820 0bc1ad162 3ce12a531		FIXED
KR-06	P2	Pad rows flow into the NAX gather; fused SwiGLU never below 4E rows	366f74339		DESIGN
KR-07	P2	Mean-based tile height; per-expert planner	4a7ee991a		DESIGN
KR-10	P2	fp16 two-pass long-Q1 partials overflow (inherited from MLX)	–		OPEN
new	–	NAX sorted gather serves affine 6-bit experts (2/3/5 exact, not admitted)	46656a8ec 7e28b1ace	35B 8K TTFT –7%	DEFAULT
new	–	GVR QSA top-k selector	223068567	synthetic only	OPT-IN

CONFIGURATION AND QUALIFICATION

ID	SEV	FINDING	COMMIT	GPU	NOW
CQ-4	P1	Default-on mechanisms the qualifier never required	734a79679 015c44843		FIXED
CQ-1	P1	Nemotron inherited Flash-Next's MTP handoff width	1eb33bb45		FIXED
CQ-2/3	P2	Lane auto refused explicit projection owners; explicit varlen contradicted	e4afc1bae 29056b2fb		FIXED
CQ-5/8	P2	Unqualified paths refused; handoff selected where it could never fire	fbbb064a5 36505df39		FIXED
CQ-6/7/9	P2	Engine vs CLI lane default; stale CLI help; mlx-mln TF32 unrecorded	575b5d972 448edec36 5d76c5340		FIXED
CQ-12	P2	Lane auto and the TensorFold tree: one projection owner	9af4bceeb		FIXED
CQ-14/15	P2	Default ownership split across five mechanisms; RoutePlanner vestigial	4ebfa5b7e		DESIGN
CQ-10	P2	Qwen3.6-35B interior checkpoints declared from sibling evidence	–		OPEN
docs	P2	Docs drift (SRPT, interior defaults, tree default, num_draft)	4377ca68a		FIXED

ADAPTERS

ID	SEV	FINDING	COMMIT	GPU	NOW
AQ-1	P0	Qwen3.5-122B crashed at server start under default lane auto	f30cce87d		FIXED
AQ-2	P1	A Qwen3.6-27B was served as Qwen3.8 with its defaults	f30cce87d 93244aa90		FIXED

ID	SEV	FINDING	COMMIT	GPU	NOW
LEAD-01/02	P1	Seven MoE families never claimed process globals; profile env edits not rolled back	e45b862e2 a79417c63 243a2f900		FIXED
G2-01	P1	Nemotron prompts rendered thinking-off by the inherited renderer	06b8a4b02 06bcb5de8		FIXED
G5-01/02	P1	Vision feature cache ignored LoRA; Gemma 4 video timestamps wrong past 16 s	3036d797e 8a8e77959		FIXED
G4-01	P1	LLaDA prompts containing the mask token broke exact routes	71b22db4f		FIXED
AQ-3..8	P2	rope_theta, 35B pad claim, 122B identity, 4B/9B budgets, Qwen3 sampling, Flash-Next compaction	d395d3c11 e45b862e2 5520c495a f7bd0f8c8 3148c73e7 9fcfc4acb		FIXED
G-series	P2	Vendor sampling, HY V3 dead MTP, DiffusionGemma argmax, Muse bf16 norm, media receipts	3148c73e7 9a52d8cf6 bedc0945a 49a04e309 a5e415400		FIXED
G2-04/G5-05	P2	Multimodal candidates need an off-pin mlx-vlm	6556bfb38		PARTIAL

CAPSULES

ID	SEV	FINDING	COMMIT	GPU	NOW
CAP-1	P1	Trajectory compiler dropped discarded movement; reported error 0.0 at 55% real error	e5514a5d0		FIXED
CAP-2/3/7	P1	Served capsule landed on row n-2 while claiming observed_used; composed capsule persisted	cd330ca82		FIXED
CAP-4/5/8	P2	Mount/bind accepted what the adapter refused later	ede895393 edc10e53c		FIXED
CAP-6	P2	Capsule requests could never restore an exact ordinary prefix	026058e1b		FIXED
CAP-9	P2	The capsule bus has no production caller	—		OPEN

REVIEW-FIX ROUND (DRIVEN BY THE POST-FIX GPU RERUNS)

ID	SEV	FINDING	COMMIT	GPU	NOW
rfix-sched	P1	External-draft and prompt-lookup prefill ran unbounded beside decode (p95 693–767 / 703–710 ms)	673aa9fe0	p95 → 463–469 / 492–508	FIXED
rfix-sched	P2	With no sample the bound took a 512-row guess (~650 ms on the 27B); stale samples never expired	7502a23ca 8c7e1fe78		FIXED
rfix-sched	P2	Max-gap outliers could not be attributed; estimator absent from route identity	d0ac480ae acadd8b76		FIXED
rfix-fn	P1	Co-arrivals left by a partial first boundary were prepared in waves (B8 TTFT 1.25 / 8.9 / 12.2 s)	057373692	B8 TTFT 6.8–7.0 → 3.0–3.6 s	FIXED

ID	SEV	FINDING	COMMIT	GPU	NOW
rfix-fn	P1	PLE prefetch re-staged the whole tail every slice (3.0 M rows staged, 0.24 M used)	daece1c6d	FN max gap 3.8–4.1 → 0.44–0.47 s	FIXED
rfix-spec	P1	PLD B4 cohort fragmented; three host syncs per verify row	1c45a38a7 3b61cbdfd 82bdcc230	plain rounds pipelined	FIXED
rfix-apc	P1	COW-1 ledger evaluated arrays: a device sync on every APCv2 store during prefill	b02360126		FIXED
rfix-apc	P2	Resident estimate counted unfreeable bytes; refused-hit withdrawal inexact; len-1 plain-hit exception too broad	b310c3e13 73548bdba 686693bee		FIXED

CODEX REVIEW FINDINGS

ID	SEV	FINDING	COMMIT	GPU	NOW
CX	P1	Hybrid abort left recurrent rows advanced; failed PLD rounds kept inconsistent lanes	1d25dd7d1		FIXED
CX	P2	Block and self-MTP verification accepted on u = h; greedy shortcut before full law validation	8ca62ebf5 91073ecad		FIXED
CX	P2	Native samplers unchecked and unserialized; partial readings fed the governor; window by record count; DOWN residency unexported	f52578e61 f03a51b6e dffad06da 01fd7a3a2 4cec3fc28		FIXED
CX	P2	Host memory floor not applied to parallel-sample admission (contradicted SERVING.md)	fc22e2885		FIXED

INTEGRATION BUGS

ID	SEV	FINDING	COMMIT	GPU	NOW
INT-1	P0	Import-order regression: the APC retention parser imported runtime.models first, so every guarded adapter refused to start	9f9a81f86	servers start	FIXED
INT-2	P0	Codex main commit 66b2f9d1e attached default composition to the tree route: DFlash2 refused at start-up	e33356767	= composition off	FIXED
INT-3	P2	Budget governor could tighten on other servers' power while its engine was idle (suspected on the GPU; the B1 pacing seen there proved correct)	f030c22fa	rerun: cap held at 24.5 W	FIXED
INT-4	P2	Preflight failures introduced by the sweep; init-bypassing test stubs lacked fairness	bac178c32 77540ea3f		FIXED

Model support matrix

What each adapter is, which route it selects, and what the sweep changed.

About twenty adapters are registered and resolved from the artifact's `model_type` and topology. The table condenses the config lane's feature × model matrix (sourced from code at 187c6aa23) and the adapter lanes' tables, updated for the sweep's changes.

STATE, FOR EVERY ROW

No model has a qualification receipt for the current source. Every descriptor carries `qualification: pending`, and any change under `src/` changes `runtime_identity()`. Read every default below as *selected*, *unqualified*; each route still serves and says so in its receipts.

MODEL	ARCHITECTURE	DEFAULT ROUTE	OTHER ROUTES	NOTABLE SELECTED DEFAULTS	CHANGED BY THE SWEEP
Qwen3.8 Flash-Next	48L: 36 GDN + 12 attention/QSA; 512-expert top-10 MoE; PLE; HC×4; MTP	native MTP K=2	ordinary · PLD	handoff >3 · copy drafts · fused GDN all phases · NAX fused · adaptive pad · QSA auto · interior auto	stall fixed (cost model, co-arrival prep, PLE staging); QSA fixes
Qwen3.8-27B	64L dense hybrid: 48 GDN + 16 attention; 1-layer MTP head	native MTP K=3	DFlash2 tree (pinned pair) · PLD · ordinary	handoff >4 · copy drafts · lane auto · fused GDN · interior auto · autoscale prefill	cohort hold; PLD batched; tree qualifiable and starts again
Qwen3.6-27B	same geometry as 3.8-27B, attested by config + template	ordinary	external continuation (unqualified)	prefill 2048 · fused GDN off	no longer served as 3.8 (AQ-2, 93244aa90)
Qwen3.6-35B-A3B	40L hybrid MoE, 256 experts top-8 + shared	native MTP if head	ordinary · DFlash2 chain · PLD	handoff 1 · NAX fused 4/6/8-bit · eager stride 2 · interior auto	6-bit NAX default; pad policy claimed
Qwen3.5-122B-A10B	48L hybrid MoE, 256 experts top-8	ordinary	—	stock kernels · NAX off	startup crash fixed; own numerics contract
Qwen3.5 9B / 4B	32L dense hybrid GDN 3:1	ordinary	PLD	engine autoscale prefill · fused GDN off	budgets at own geometry
Nemotron 3 Super / 3.5 Lightning	Mamba2 + GQA + latent MoE	ordinary	native MTP (opt)	prefill 2048	own renderer; handoff not inherited
gpt-oss (+ Puzzle)	alternating SWA-128/full, sinks, mxfp4 MoE	ordinary	—	harmony channels · thinking guard (Puzzle)	MoE claims; dead hooks removed
Gemma 4 31B / 26B-A4B	via pinned mlx-vlm with mlx2 caches	ordinary	—	projected-feature reuse · prefill 2048/512	LoRA-bound feature cache; video times
Xing4.0	DSv3 MLA + sigmoid MoE + mHC + MTP	native MTP K=1	ordinary · PLD	handoff 4 · mHC kernel	MoE claims; env rollback
Muse Glimmer	local mlx2 vision tower	ordinary	DFlash2 chain (unqualified)	prefill 2048 · composition on chain (unmeasured)	fp32 centered norm
North Mini Code	Cohere2-MoE parallel block, sigmoid top-8	ordinary	EAGLE · PLD	thinking budget 512 · composition on chain (unmeasured)	receipt from charged transient

MODEL	ARCHITECTURE	DEFAULT ROUTE	OTHER ROUTES	NOTABLE SELECTED DEFAULTS	CHANGED BY THE SWEEP
Laguna XS / S	1:3 full/SWA-512, partial YaRN	ordinary	DFlash k=3 (XS)	fused MoE off (slower)	MoE claims
Granite SWA · HY V3 · Agnes	MoE SWA · Hy3 MoE · GDN hybrid	ordinary	—	vendor sampling now declared	HY V3 dead MTP removed
LFM2.5-VL · SmoIVLM2 · Qwen2.5-VL	multimodal bridges (off-pin mlx-vlm)	ordinary	—	serve labelled unqualified	clear revision refusal
LLaDA · DiffusionGemma	diffusion LMs, direct adapters	direct	—	seeded; reference sampling	mask-token refusal; multinomial

Cross-cutting defaults

MECHANISM	WHERE IT IS SELECTED	STATE
Decode-time fairness (500 ms target)	every ordinary and MTP route (fair share 0.5); external-draft and PLD routes since 673aa9fe0 (fair share 0)	DEFAULT cost model; identity changed
Lane matmul <code>auto</code>	dense M5 routes (Qwen3.8/3.6-27B, 4B/9B, Gemma 4 31B, LFM); off for MoE	DEFAULT engine matches CLI (575b5d972)
Interior checkpoints <code>auto</code>	Flash-Next, Qwen3.8-27B, Qwen3.6-35B native-MTP routes	DEFAULT 35B unmeasured (CQ-10)
SRPT prefill order	native MTP with ≥ 4 lanes	DEFAULT steps aside below 4
MoE NAX sorted gather widths	MoE routes with NAX on: affine 4/6/8-bit + MXFP4	DEFAULT 6-bit new, bit-identical; needs requal
Proposal composition	external chain-route drafters with draft distributions (not tree routes)	DEFAULT from main 66b2f9d1e; unmeasured
Floor-3 sorted-MoE rhs pad	every MoE family on the floor policy	BIT-CHANGING unqualified, now in receipts
Power telemetry · governor · keep-warm · value retention · block verify · GVR selector	flags / policy keys	OPT-IN new this sweep

Sources: `qualification/runs/sweep-20261006/lanes/config-qual.md` (Deliverable 1), `adapters-qwen.md`, `adapters-other.md`; `src/mlx2/adapters/registry.py`; commits named.

Honest limits and open decisions

What a reader should not conclude from this document, and what needs a decision.

This chapter collects what a reader could otherwise over-read from the earlier ones. Nothing here is hidden elsewhere; it is gathered so a decision-maker sees it in one place.

What is not true yet

- **No route is qualified at the current source.** The sweep changed `src/`, so every earlier receipt is void. Several changes alter selected behaviour and need requalification before any route can claim a receipt: the cost-model estimator (now in route identity), the 6-bit NAX default, decode fairness on external-draft and prompt-lookup routes, the composition default on chain routes, and harness re-pins.
- **Prompt lookup B4 reached parity late in the sweep:** 55–63 tok/s against ordinary 62–63 on the same run (was 25–33), after the cohort polls all verified tokens, reuses its batched views and pipelines plain rounds (82bdcc230; final-pld). B1 still costs about 10% over ordinary.
- **Max-gap outliers remain.** Every route's p95 neighbour gap is at or under about 500 ms, but maxima reach about 1 s on native MTP, 0.8 s on prompt lookup and 0.64 s on DFlash2. The new gap attribution counters can now say why; nobody has read them on the GPU yet.
- **Warm hit depth is unstable** on native MTP (27B: 4,096 vs 7,168 tokens reused across reps) and on Flash-Next ordinary (8,192 of 16K once). Not yet explained.
- **External DFlash2 has no shared-prefix reuse, by design** (no paired draft state below the prompt boundary): warm 8K TTFT about 8.6 s against 1.6 s on ordinary. Lifting it needs draft-state checkpoints.
- **Test suite:** the full CPU suite passes on the M5 at the final tree (11,646 tests, 42/42 import guards). The M3 runs more failures only where it lacks `mlx_vlm`, native binaries or the TensorFlow checkout; one stale test on main was fixed (cce60ef5c).
- **The floor-3 MoE rhs pad is on by default, changes bits and is unqualified;** receipts name it. Keeping it or turning it off until qualified is an open decision.
- **Measured, and therefore off:** GPU keep-warm saves about 0.3 s after 20 s idle (stays opt-in). **Unmeasured:** block verification's gain, the GVR selector end to end, composition on chain routes. Per-request joules are an allocation estimate, not a meter.
- **The governor is a cap, not an efficiency mode.** Budget mode held 25 W, at about –67% throughput and +15% J/token; `efficient` saved only about 2% J/token on the measured workload. `burst_reserve` (spend the 90 s burst on interactive TTFT) is not implemented.
- **The capsule bus has no production caller** (CAP-9); activation capsules are reachable only from tests and scripts.
- **Deployment drift.** At baseline the live 27B service ran a policy from an unmerged worktree (compiled replay) that main's adapter refuses; the repo default and the service disagreed.

Fixed since the first build of this review: the opt-in host memory floor now also applies to parallel-sample ($n > 1$) admission, as `docs/SERVING.md` states (fc22e2885); telemetry overhead and keep-warm were measured (chapters 9 and 4).

Design items the sweep left open

ITEM	WHY OPEN
Near-tie contract (SPEC-02)	It cannot fail on rate or cross-arm shift, so a small systematic perturbation passes as near ties. Project rule: near ties are not failures; whether to add a rate gate is a policy call.
Composition gate (SPEC-04 remainder)	Index bounded to the lookback window; no acceptance gate or rejected-source feedback yet; now a default on chain routes without a measurement.
Native continuations (SS-9)	Default-off research routes; would need counting in active-decode and timing into the cost model.
NAX pad rows (KR-06)	Skipping pad rows and admitting fused SwiGLU below 4E needs a GPU bit-identity A/B; plan written (366f74339).
Declared-defaults table (CQ-14/15)	Proposal only (4ebfa5b7e): one per-adapter table, data-driven qualifier lists.
Governor <code>burst_reserve</code>	Needs request priority classes in admission, which the scheduler does not have.

Research verdicts that close doors

- **Histogram MoE tile planner: do not build.** The mean-based plan is within 0.5–0.7% of a per-call best plan on real 35B routing; the 6-bit NAX path it pointed to is now built.
- **Routed A8 gate+up: approximate.** Gate+up is 24.8% of 35B prefill, so 1.3–1.5× would give +6–9% across all layers or +2.4–3.5% on the NAX layers only; it changes numerics and would need its own qualification. Not applicable to the 27B, whose adapter declares no int8 prefill scope.
- **GDN chunk16 prefill: do not select.** 1.19–1.46× per layer at 27B shapes, but inexact (state error ~4e-3, 0.8–1.6% argmax flips) for about 1% of prefill; end to end within thermal noise.

Decisions for the maintainer

1. Requalification order for the changed defaults (27B routes, Flash-Next served policy, 35B with 6-bit NAX), and whether the external routes' new fairness settles before or after the PLD plain-round fix lands.
2. Composition on chain routes: keep the default from main without a measurement, or require an A/B on Muse DFlash2 and North EAGLE first.
3. Floor-3 MoE pad: keep as an unqualified default, or set the module default to 0 until a campaign qualifies it.
4. Slice floor for decode fairness (1024 gave TTFT –17%–20% but neighbour gap +55% and lane tok/s –9% in the 10-02 sweep).
5. Whether to adopt the declared-defaults proposal before the next qualification campaign; re-pinning mlx-vlm for the multimodal candidates.

Evidence and method

Where every number comes from, and how to rebuild this document.

Campaigns

Before/after charts compare **before** with **final** only. Intermediate campaigns ran on the integration tree before the review fixes and are used only for A/Bs whose arms ran identical code (estimator, keep-warm, telemetry overhead, governor modes) and to show the Flash-Next fix in stages; `post-q38-routes` is superseded by `final-q38-routes`. The intermediate tree also carried the COW-1 ledger's device sync (fixed in b02360126), one more reason not to compare its absolute numbers with the final ones.

STAGE	CAMPAIGN	MODEL	ARMS	WORKLOADS	REPS
Before origin/main 187c6aa23	<code>q38-routes</code>	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash, pld	short, long:8000, warm:8000, batch:4, contention:12000, copy, json, tool	2
	<code>fn</code>	Qwen3.8-Flash-Next-Uncensored-MLX2-4bit-MTP	served, ordinary, served_qsa_direct4	short, long:32000, warm:16000, batch:8, contention:16000, copy, json, tool	2
	<code>q38-features</code>	Qwen3.8-27B-oQ4e-mtp	served, lm_off, no_interior, int8_mlp, gdn_chunk16	short, long:8000, warm:8000, batch:4, contention:12000	2
	<code>q36</code>	Qwen3.6-35B-A3B-Abliterated-Heretic-MLX-4bit	mtp, ordinary	short, long:16000, warm:8000, batch:8, contention:12000, copy, json, tool	2
	<code>thermal-q38</code>	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash	thermal map: prefill, B1, B4 segments	25 min hot
Intermediate integration tree 816aef02c (identical-code A/Bs only)	<code>estimator</code>	Qwen3.8-27B-oQ4e-mtp	mtp_cost, mtp_runmax, ord_cost, ord_runmax	contention:12000, batch:4	2
	<code>keepwarm</code>	Qwen3.8-27B-oQ4e-mtp	off, keepwarm	idle:2, idle:6, idle:20	3
	<code>telemetry-overhead</code>	Qwen3.8-27B-oQ4e-mtp	off, telemetry	short, batch:4	3
	<code>governor</code>	Qwen3.8-27B-oQ4e-mtp	maxtp, efficient, budget25, budget15	thermal map: prefill, B1, B4 segments	20 min hot
	<code>post-q38-routes</code>	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash, pld	short, long:8000, warm:8000, batch:4, contention:12000, copy, json, tool	2

STAGE	CAMPAIGN	MODEL	ARMS	WORKLOADS	REPS
	post-fn	Qwen3.8-Flash-Next-Uncensored-MLX2-4bit-MTP	served, ordinary	short, warm:16000, batch:8, contention:16000, copy	2
Final main 43d561e38	final-q38-routes	Qwen3.8-27B-oQ4e-mtp	ordinary, mtp, dflash, pld	short, long:8000, warm:8000, batch:4, contention:12000, copy, json, tool	2
	final-fn	Qwen3.8-Flash-Next-Uncensored-MLX2-4bit-MTP	served, ordinary	short, warm:16000, batch:8, contention:16000, copy	2
	final-q36-6bit	Qwen3.6-35B-A3B-Abliterated-Heretic-MLX-4bit	nax48, nax468	short, long:8000, long:16000, batch:8	3
	final-governor	Qwen3.8-27B-oQ4e-mtp	maxtp, budget25	thermal map: prefill, B1, B4 segments	15 min hot
	dflash-comp	Qwen3.8-27B-oQ4e-mtp	dflash_default, dflash_nocomp	short, long:8000, batch:4, copy, json	2

WORKLOAD DEFINITIONS (SCRIPTS/PROFILE_FEATURES.PY)

- **short:** one short prompt; decode tok/s = (tokens - 1) / (last - first token arrival).
- **long:N:** a fresh N-token document, three-bullet question; TTFT is the cold prefill.
- **warm:N:** the same document twice with different questions; the second request's TTFT and cached tokens.
- **batch:B:** B concurrent 700-token notes. This review reports batch wall time (tokens ÷ first submission → last token) for Flash-Next B8; the profiler's aggregate (tokens ÷ first token → last token) is shown for the 27B B4, where starts are not staggered enough to matter.
- **contention:N:** a long decode starts, 1.5 s later an N-token prompt arrives; the decoding request's gap p95 and max over the prefill window (neighbour block).
- **idle:S:** a 4K cold-prefix prompt after S seconds idle (keep-warm).
- **copy / json / tool:** verbatim reproduction of 450 tokens; a JSON-schema task record; one function call.
- **energy:** IOReport GPU + DRAM joules over the workload ÷ output tokens; P-state residency and die temperature bracket each workload.

PRIMARY SOURCES

qualification/runs/sweep-20261006/FINDINGS-GPU.md (GPU ledger, authoritative interpretations, later sections supersede earlier ones) · lanes/*.md (lane reports, history digest) · qualification/runs/research-20261006/ · docs/SERVING.md, docs/METRICS.md, docs/QUALIFICATION.md, docs/PROVENANCE.md · git log origin/main...HEAD. Mechanism constants were read from source at the review tree; file names are given in each caption.

CORRECTIONS CARRIED FROM THE LEDGER

The baseline's apparent route prefill penalty was heat (first run on a cold machine). The Flash-Next B8 "regression" was a metric artifact; batch wall time is used instead. The governor's B1 throttling under a 25 W budget is correct, because the budget covers GPU + DRAM.

REBUILDING THIS DOCUMENT

python3 docs/architecture-review/build.py — standard library only; headless Chrome renders the PDF and poppler's pdftotext fills the contents page numbers. A missing campaign renders as a dashed pending panel. Built from 43d561e38.